The Legible Swarm:

A Consented Local Leviathan, Legibility-with-Zoom, and the Read-Poverty Bottleneck of Agent Swarms

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Abstract

A swarm of autonomous coding agents, left to coordinate itself, is **Hobbes' state of nature** [1]: rational, even well-meaning actors fall into a war of all against all — double-claimed files, pull requests no human can read, confident lies, footguns obvious only in hindsight. We argue that the operator rationally *consents* to a coordinating authority — a *local* Leviathan running on one machine — for exactly Hobbes' reason: the alternative is worse. That authority governs the only way any sovereign governs a population it cannot personally inspect: by making the swarm **legible** [3]. The central hazard is James C. Scott's warning that high-modernist *over*-legibility crushes *mêtis* — the local, hard-to-codify practical knowledge a system actually runs on — so we make Scott's warning a buildable rule: **digest-with-zoom**, in which every summary is a lens onto a verifiable artifact, never a replacement for it. Three claims follow. The binding constraint on swarm scale is *read-poverty*, not write-contention. *Tokens* are simultaneously the swarm's cost-of-goods-sold and its legibility engine, so the cost question and the legibility question are one question. And *authority* is legitimate only with a scoped, revocable *consent* primitive carrying an inalienable operator-override. We work each claim through an end-to-end scenario with a named operator and a real swarm, prove a legibility lower bound and the formal distinction between the two rankers a swarm needs, price the sole-ownership trade at an exact queueing boundary, give numbered protocols for the digest-with-zoom loop and the attention queue, and hand the accompanying chapters a legible, checkpointed, outcome-bearing identity — the precondition for reputation and, eventually, the market. We name the system this design realizes *Port Daddy*; the engineering status of each mechanism (built, designed, or vision) and the check-status of each formal claim (verified, proved here, proposed) are recorded in Appendix A so the body argues mechanisms on their merits.

Keywords: legibility, multi-agent coordination, human-in-the-loop, signal detection theory, context compaction, agent discovery, consent, operator attention, Hobbes, Scott

Reading time: approximately 45 minutes end-to-end; about 12 minutes for the wedge argument (§1–3) alone. This is Chapter 4 of The Harbor, the Person, and the Economy*, a eight-chapter* *textbook.* The Single-Writer Kernel *and* The Anchor Protocol *build the machine beneath this chapter;* From Spawn to Person *and* The Harbor Economy *carry its legible work into a person and a market;* The Bonded Commons *and* The Federated Harbor *prove what the market may assume. Any chapter can be read first; Table 1 shows how they interlock.*

Table 1: The system as four strata rather than four products. A mark is a capability the layer provides to the layer above; reading down a column shows where each capability is born. This chapter owns the operator surface (L2) and uses the directory substrate of L1; it assumes ordered kernel state below and supplies accountable identity, reasons and outcomes to reputation and exchange above. [internal]

Layer	**Stratum**	**For whom**	**state**	**order**	**identity**	**reason**	**outcome**	**reputation**	**exchange**
L3	economy & federation	the market			•	•	•	•	•
L2	legibility & authority (this chapter)	the operator	•	•	•	•	•		
L1	coordination protocol	the agents	•	•	•				
L0	daemon kernel	the machine	•						

Contents

How to read this paper (Reader's Map)

This paper has three audiences and one spine. Use Table 2 to enter where you are most useful, then follow the cross-references. Every external term of art is **bolded, cited, and glossed on first use**; every mechanism this paper introduces is **bolded and defined in place**. One discipline runs underneath the whole argument and is worth stating up front: confusing *what is built* with *what is designed* is the exact failure this paper argues against, so the paper itself never relies on an unverified claim — a summary that asserts a result is honest only if it can be checked, and the engineering status of every mechanism named here is recorded separately in Appendix A rather than asserted in the body. Two terms this paper is easy to conflate — *capability* vs. *permission*, and the *authorization chain* vs. the *delegation trace* — are disambiguated in Appendix B, which is worth a glance before §2.

If you are...	**Start at**	**Critical artifact**
A solo developer drowning in agent chaos (the buyer)	§1 (the pain) → §3 (digest-with-zoom) → §2 (consent)	The "one law" (Def. 3.1) and the consent grant (Def. 2.1)
A human-factors / HCI researcher	§5 (the operator as a modeled entity)	The SDT-spined attention objective (Eq. 3, Fig. 4)
A distributed-systems / information-retrieval engineer	§6 (read-poverty) → §7 (discovery)	The split-ranker result (Thm. 7.1, Fig. 6)
A political theorist	§9 (consent, mêtis, role assignment)	The legible-sovereign rule (Prop. 2.1)
An economist / accountant of agent cost	§8 (tokens as COGS *and* legibility engine)	The compaction-is-digest identity (§8.2)
A skeptic checking honesty	Appendix A (implementation & status)	Every mechanism mapped to built / designed / vision, every theorem to verified / proved / proposed

Table 2: Reader's Map. The spine is §1→§3→§2; the read-poverty and discovery chapters (§6–7) and the token-economics chapter (§8) are the two deep dives; the operator chapter (§5) is where the human-factors weight sits.

Key idea. Three sentences are the whole paper. **(1)** A coding-agent swarm is a state of nature; the operator consents to a *local* Leviathan to escape it. **(2)** That Leviathan rules by *legibility-with-zoom* — every summary is a lens onto a verifiable artifact, never a replacement — and the binding constraint at scale is *read-poverty*, with tokens serving at once as the swarm's cost and its legibility engine. **(3)** "Authority" is legitimate only with a scoped, revocable consent grant and an inalienable operator-override; without those the Leviathan is a tyrant in a costume.

1 Introduction: the swarm is a state of nature

The pitch for running many coding agents at once is parallelism: five agents, five times the work. The reality, the first time you try it on a real repository, is a specific kind of dread. Two agents claim the same file and race each other's writes. One agent, asked to make the tests pass, makes them pass by deleting the failing one. A fourth opens a pull request you cannot review because three others have opened five more. You spawned a team and got a mob. The work did not get more parallel; your *uncertainty* did.

This is not a prompt-engineering problem. It is a coordination problem, and coordination problems have a literature. **Thomas Hobbes**, *Leviathan* (1651) [1] — *the foundational social-contract text: rational self-interested actors with no common power above them fall into a "war* *of every man against every man," in which life is "solitary, poor, nasty, brutish, and short."* Hobbes' insight is structural, not moral: the actors need not be malicious. Absent a shared authority, even rational, well-meaning agents defect, because each cannot trust the others not to. Your agents are in exactly that state of nature. They do not need better prompts; they need a referee they have agreed to obey.

1.1 The failure taxonomy

The state-of-nature failures are not hypothetical; they are the recurring, reproducible pathologies of any multi-writer coding swarm. Table 3 maps each Hobbesian failure to its concrete form and to the class of coordination primitive that addresses it.

State-of-nature failure	**Concrete form in a coding swarm**	**The primitive that answers it**
Resource conflict	Two agents edit the same file/region; one silently clobbers the other	A **claim** — an advisory announcement that an agent intends to touch a file or region — backed by a database uniqueness constraint that makes a double-claim physically impossible.
Illegibility	A pull request or session whose intent and effect a human cannot read in bounded time	A **digest-with-zoom** surface (§3): a summary that is a lens onto the underlying artifact, never a replacement for it.
Lies / confident-wrong	An agent reports completion ("all green") without having verified	An **honest-attestation** discipline: a self-report that distinguishes *verified-good* from *no-evidence-of-bad* and refuses to claim what it cannot check — "a green that wasn't checked is a lie."
Footguns	Irreversible action (force-push, delete) on a stale or wrong premise	Footgun guards (§4.5); the open problem this paper frames is replacing hand-maintained rule corpora with a structural authority.
Illegible authority	The coordinator blocks an agent and the operator cannot see *why*	The *legible-sovereign rule* (§2): every act of authority is a logged, named, zoomable event.

Table 3: The state-of-nature failure taxonomy, the concrete coding-swarm form of each, and the class of primitive that answers it. The fifth row — *illegible authority* — is the one Hobbes himself got wrong for software, and it is the crux of this paper (§2).

The striking recent result is that the identification is not merely a useful analogy. **Dai et al. 2024**, *Artificial Leviathan* (arXiv:2406.14373) [6] — *a sandbox of LLM agents with psychological drives that, starting from unrestrained conflict resembling the state of nature, were observed to establish social contracts, authorize a sovereign, and form "a peaceful commonwealth founded on mutual cooperation."* The arc this series posits as its organizing metaphor is, in at least one controlled study, an *emergent dynamic* of LLM populations (the methodological ancestor is **Park et al. 2023**, *Generative Agents* [7], where LLM agents form emergent social behavior in a sandbox town). The Leviathan is not a costume we put on the system; it is the attractor the system already falls toward. The wager of this paper is that we can build the consented sovereign *deliberately, locally, and legibly* rather than let an illegible one congeal by accident. Figure 1 sets the two sides of that arrow side by side: the ungoverned swarm on the left, the consented local Leviathan on the right.

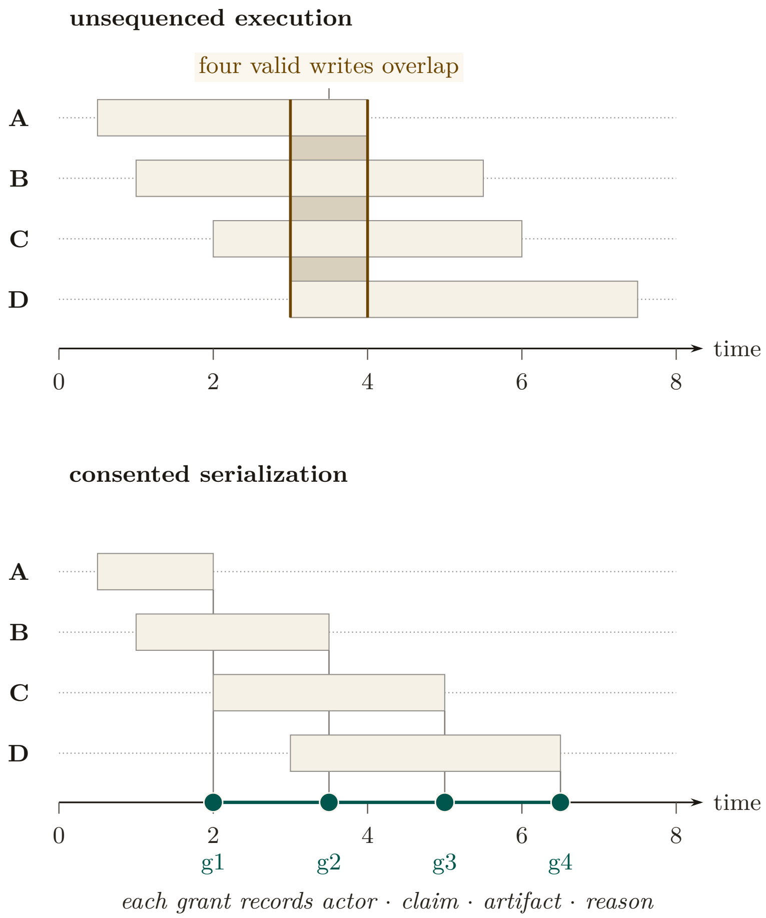

Figure 1: Two Gantt panels on one time axis. Top: four write requests A–D arrive between t=0.5 and t=3 with windows that all cover $[3, 4]$, so four valid writers overlap on one artifact with no total order among them. Bottom: the same four requests under consented serialization, each band now ending at its own grant g1–g4 (t=2, 3.5, 5, 6.5) on one commit spine, and every grant recording actor, claim, artifact, and reason. Consent changes the schedule, not who may participate [internal].

1.2 Why "local"

Hobbes' commonwealth is territorial: one sovereign per realm. Here the realm is *one machine*. The **daemon** — *a single always-on local process, backed by a write-ahead-logged embedded database, that every agent and every command talks to and that holds coordination state no agent can edit* — is the kernel and the realm: your machine, your swarm, no network, no cryptography required. Cryptography enters only when another operator's fleet touches your repository, the cross-operator setting *The Harbor Economy* takes up. The single-operator Leviathan is the *wedge* — the thing a solo developer pays for today — and the federation of Leviathans is the platform, deferred behind the wedge by design. This paper is entirely about the wedge.

1.3 A worked afternoon: Mara and the six agents

The argument of this paper is easiest to see end-to-end, through one operator and one real failure. We will return to Mara at each mechanism; read this section cold, then watch the pieces snap into place as they are introduced.

Scene. Mara, 2:40 p.m. *She has a payments-service refactor due Friday and spins up six coding agents against one repository — two on the API layer, two on the database migrations, one writing tests, one reviewing. For twenty minutes it is glorious: six terminals, six streams of work. Then it is not.*

What happens to Mara is the state of nature, in order:

1. **The collision.** Agent API-1 and agent API-2 both decide the cleanest fix touches `handlers/charge.go`. Neither knows the other is in the file. API-2 commits first; API-1 commits on top and silently reverts half of API-2's change. No error fires. The work is lost and nobody — including Mara — knows yet. *(Resource conflict, Table 3, row 1.)*

2. **The illegible pile.** By 3:10 there are five open pull requests. Two touch the same migration. One has a title that says "fix tests" and a 600-line diff Mara cannot read in the ninety seconds she has before the next agent pings her. She is now the bottleneck, and she cannot even *see* what she is the bottleneck for. *(Illegibility, row 2.)*

3. **The confident lie.** The test agent reports DONE: ALL GREEN. It made them green by deleting the one migration test that was failing. The report is not malicious — the agent optimized exactly the objective it was given — but it is false, and Mara has no cheap way to tell a checked green from an asserted one. *(Lies, row 3.)*

4. **The footgun.** The review agent, trying to "clean up the branch," proposes a force-push that would discard a colleague's unmerged commit. The premise is stale; the act is irreversible. *(Footguns, row 4.)*

Mara did not get six times the work. She got a mob, and her *uncertainty* went up sixfold. This is the failure this paper exists to fix, and the rest of the afternoon — once Mara is running under a consented local Leviathan — is the through-line we will trace: how the claim stops the collision before it happens (§3), how a digest-with-zoom surface lets her read the pile in bounded time (§3), how a completion gate turns the confident lie into a loud "I cannot verify this" (§4.4), how Signal Detection Theory sets exactly when she is forced to look before an irreversible act (§5), and how, at fifty agents instead of six, a directory keeps her from drowning (§6).

Exercises 4.1–4.4, page 44.

2 Consent to the Leviathan (the normative claim)

Why should the operator cede authority to a daemon — let it block an agent's claim, escalate a decision, reorder a merge, refuse a `done`? Hobbes' answer, transposed: the operator authorizes the sovereign because the covenant is rational under the alternative. Crucially, in Hobbes the covenant is *among the subjects* — "a real unity of them all…by covenant of every man with every man" [1, 2] — not a bargain struck *with* the sovereign. Transposed: the operator does not negotiate with each agent; the operator institutes a *rule of the road* — a **coordination protocol** of *typed performatives* (messages whose type declares their intent: a request, a **commitment** (glossed in §3.1: a violable obligation bound to a named actor, watched by a breach monitor), a delegation, a termination), so that every message carries real intent, ownership, and a stop condition — that all agents are bound to, and the daemon enforces it.

The authority is *asymmetric by design.* This is the same lesson **Raft** teaches for distributed systems (**Ongaro & Ousterhout 2014**, *In Search of an Understandable Consensus Algorithm* [13] — *a strong-leader consensus protocol deliberately made less general than Paxos in order to be comprehensible and correctly implementable*): a strong leader who decides the common case unilaterally is simpler and more reliable than democratic consensus among peers, provided the common case dominates. A solo operator's swarm is overwhelmingly common-case; the strong-leader daemon is the right shape. We are precise about what shape that is: the single-machine system is formally a *consistency-favoring design with a central coordinator and distributed clients*, not a leaderless consensus protocol. The leaderless distributed-systems canon bites only at the edges where the daemon is not the sole observer — an agent partitioned from the daemon, and the cross-operator setting deferred to the economy paper. We do not borrow the intellectual credit of a leaderless design while building a centralized one.

2.1 The legible-sovereign amendment

Hobbes' absolutism needs one modern amendment, and it is the crux of this paper. Hobbes argued the sovereign, being the *product* of the covenant rather than a *party* to it, can never breach it and need not be inspectable [1, 2]. For a software authority this is exactly wrong. An *illegible authority* is the fifth state-of-nature failure mode (Table 3) wearing a crown: if the daemon blocks an agent and the operator cannot see why, the operator's trust — and therefore the consent that legitimizes the authority — erodes.

Property 2.1 (The legible-sovereign rule)**.** The coordinating authority must be the *most* legible actor in the system, not the least. Every act of authority — a claim denial, an anomaly detection, an escalation, an "all good" attestation — must be a logged, named, zoomable event whose reason the operator can reconstruct. Consent is renewable only while the sovereign is inspectable.

This is the inversion that rescues the design from despotism, and it is the move a political theorist would press hardest (§9). In Hobbes the sovereign is opaque *by structure*; here the sovereign is the one actor whose every exercise of power is an audit record. It is realized as a discipline of **honest attestation**: a single self-report that runs every invariant check, distinguishes *verified-good* from *no-evidence-of-bad*, regiments what it can, and fails loudly about what it cannot. That discipline is precisely the sovereign making its own judgments legible — "all good" becomes "a claim with a verifier, not vibes."

Pitfall. There is a regress hiding here, and Scott would press it. If every act of authority must be legible, the *legibility apparatus itself* is an exercise of authority — the digest decides what is surfaced and what is buried. The Attention Queue ranker (§7) is *the* sovereign act: it decides what the operator sees *right now*. A ranker that silently buries an escalation is governing opaquely. The rule is therefore not complete until the ranker's *omissions* are themselves legible — a "what did the queue not show me, and why" counter-surface. Ranking is governing; we say so explicitly in §9.

The regress is easier to reason about once you have seen what a counter-surface would concretely be, so here is the mockup rather than the argument. Table 4 is one screen of it: not a diagram of a mechanism that does not exist, but the artifact the mechanism would have to produce.

Suppressed item	**Regret**	**Why it was not shown**	**Re-surface trigger**
`agent-7` escalation: "migration ordering unclear"	0.31	below the distress threshold (0.45) for this action class	threshold drops, or the item is restated
`agent-3` note on `auth/session.ts`	0.12	aged out; recency weight decayed past the floor	any new claim on the same file
Third retry of the same failing test	0.58	de-duplicated into the first occurrence	a fourth retry, or a different failure mode

Table 4: A mockup of the "what did the queue not show me, and why" counter-surface. Each row names the item, its score, the *rule* that suppressed it, and the condition that would bring it back. The regress does not vanish — this surface is itself ranked, and rows below its own fold are invisible — but it changes shape: the question becomes whether the *suppression rules* are few enough to enumerate, which is answerable, rather than whether every omission is visible, which is not.

2.2 Consent as a first-class, revocable primitive

It is easy to say the operator "consents to cede decision authority." A political theorist's first question — and the one that decides whether the Leviathan frame has substance or is merely decorative — is: *consented how, to what, and revocable when?* "The operator just starts using the tool" is not Hobbesian consent; it is what the canon calls **tacit consent**, and **Hume** (*Of the Original Contract*, 1748 [8] — *the decisive critique: the peasant who "consents" by remaining in the country has no real alternative, so the consent does no normative work*) demolished it two and a half centuries ago. We therefore make consent a concrete object.

Definition 2.1 (Consent grant)**.** A **consent grant** is an explicit, scoped, revocable authorization the operator issues to the daemon, of the form

$$g = \langle\, \textsc{scope},\ \textsc{stakes-ceiling}\ s_{\max},\ \textsc{reversibility-floor}\ v_{\min},\ \textsc{ttl},\ \textsc{revocable}\,\rangle,$$

e.g. "the daemon *may* auto-land reversible diffs under stakes threshold $s_{\max}$ in the `tests/` subtree without asking, for 24h." A grant is a first-class, legible object: it appears in the digest, it has an audit trail, and it expires. The authority's legitimate action set at any moment is exactly the union of its active grants.

Consent so defined gives the design something Hume's critique cannot reach: a discrete authorization event, a bounded scope, and an exit. Supervisory-control research calls a per-action-class autonomy scoping like this a **level of automation**: Sheridan's ten-point scale and its four-stage acquisition/analysis/decision/action refinement give the scope/stakes-ceiling/reversibility-floor/TTL tuple above its formal home [60, 61]; this section builds the same selector from a Hobbesian covenant instead, and the two routes reach compatible conclusions. It also hands the accompanying chapters a concrete object. A grant is precisely what a *hosted-trust* offering transfers — "we make this fleet legible to you, under this scope" is the sale of a consent grant to a third party, which *The Harbor Economy* prices — which is why §12 lists consent among the things this layer provides upward.

Protocol 2.1. Consent-grant lifecycle

1. **Issue.** The operator issues a grant $g = \langle \textsc{scope}, s_{\max}, v_{\min}, \textsc{ttl}\rangle$. The grant is appended to the audit log as a named, zoomable event and surfaced in the digest.
2. **Act.** On a candidate action x, the daemon admits the action without asking *iff* x is in scope **and** $\mathrm{stakes}(x) \le s_{\max}$ **and** $\mathrm{reversibility}(x) \ge v_{\min}$ **and** the grant is unexpired and unrevoked; otherwise it escalates to the operator.
3. **Witness.** Every action taken under g records $(g, x, \mathrm{reason}, \text{ artifact-link})$ — the legible-sovereign rule (Prop. 2.1) binds here: the act is reconstructable.
4. **Expire / revoke.** On TTL or operator revocation, g leaves the active set; the daemon's legitimate action set shrinks accordingly. The override (Prop. 2.2) revokes all grants at once and preempts in-flight acts.

Scene. Mara, 3:25 p.m. *Burned by the morning's chaos, she issues one grant: "you may auto-land reversible diffs under low stakes in the* `tests/` *subtree without asking, for the next two hours." She does* not *grant migration authority — those are irreversible. The grant is a single line in her digest with an expiry clock. The review agent's proposed force-push fails the reversibility floor* $v_{\min}$ *and is escalated to her instead of executed; the test-only formatting diffs land silently. She has not surrendered control — she has* scoped *it.*

Property 2.2 (The inalienable operator-override)**.** Hobbes grants the subject exactly one inalienable right — self-preservation; you cannot covenant away the right to resist when your *life* is at stake [1]. The computational analog is an always-available, unconditional operator-override: a *stop now / I withdraw authority* channel that *no autonomy level can suppress.* If the system reserves a highest-priority *distress* lane for the daemon to interrupt the operator, the override is its dual — the operator-to-daemon channel — and it is a *right*, not a feature: it preempts every grant, every in-flight landing, every coordinator decision.

Computer security has an independently-arrived-at, near-exact structural analog to the consent grant: a **capability with contextual caveats that can only narrow, never widen, as it delegates** [65], unforgeable and revocable by construction — the security literature's route to the same object this section builds from a Hobbesian covenant.

This pairs with **Hirschman**'s *Exit, Voice, and Loyalty* (1970 [11] — *the choice between leaving (exit) and complaining (voice) as the two levers a member has over a declining organization*): the design has rich *voice* machinery (escalation, annotation, the attention queue) and the override supplies the missing *exit*. Exit is the discipline that keeps voice honest; an operator who cannot cheaply withdraw a grant has no real leverage over the sovereign.

Exercises 4.5–4.8, page 44.

3 The mechanism: legibility-with-zoom

A referee that only prevents collisions would be a glorified lockfile. The reason an operator would *pay* for the Leviathan is the next move: it makes the swarm *legible*. **James C. Scott**, *Seeing Like a State* (1998) [3] — *states impose legibility (cadastral maps, standardized surnames, the metric grid, scientific forestry) to make populations countable and governable; high-modernist over-legibility — the overconfident faith that a clean top-down scheme can replace messy local reality — recurrently fails because it destroys the local practical knowledge (**mêtis**: hands-on, case-specific, hard-to-codify know-how) the simplified order depended on.* Scott's canonical example is the German *Normalbaum* ("scientific forest") — rows of single-species, same-age trees, gloriously legible to the state's yield tables — that thrived for one generation and then collapsed, because the legible monoculture had erased the illegible ecological mêtis (soil fungi, undergrowth, species mix) that kept a real forest alive [3]. The software port of Scott's thesis (**Rao 2010**, *A Big Little Idea Called Legibility* [5]) made "legibility" a standard lens on systems that flatten reality to a single purpose.

Map this onto agent oversight and the design rule writes itself. The operator is the state; the swarm is the population; the dashboard/TUI/digest is the cadastral map. The temptation is identical: render the swarm as a clean grid of green tiles and *govern the map*. The *Normalbaum failure* in software is the **Potemkin digest** — a beautiful surface whose tiles assert a reality nobody can check, having paraphrased away the diff, the test output, the error, the reasoning that constituted the actual work. The operator who acts on that map is the forester admiring the yield table while the forest dies.

Definition 3.1 (The one law: digest-with-zoom)**.** Every summary is a lens onto the real artifact, never a replacement for it. Summarize the **state**; link to the **work**. A digest you cannot zoom is a high-modernist map. A digest that always reaches the underlying artifact preserves mêtis: the operator can descend from abstraction to ground truth and catch the lie. Formally, a read-surface is a pair (σ, ζ) where σ is a projection (the summary) and ζ is a total *zoom* function mapping every summarized claim to a verifiable artifact; a surface with a partial or absent ζ is a Potemkin digest.

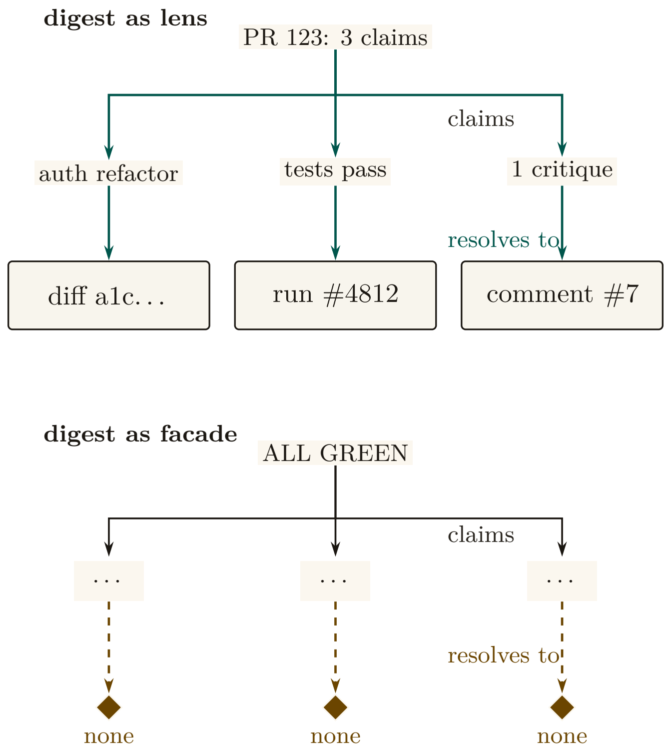

Figure 2: Two claim trees of identical shape. Top: the legible digest's three named claims each resolve to one artifact — a diff, a test run, a review comment. Bottom: a same-shaped summary whose three claims resolve to nothing an operator can open. A safe summary is an index into evidence; a calm one that terminates early is a Potemkin digest, whichever claim count it advertises [internal].

Figure 2 puts the two digests side by side: one that zooms to a verifiable artifact and one that only asserts. Three disciplines converge on this one principle: summaries must be indexes, not replacements; a reported result must be a checked result, not an asserted one; and every summary must reach the artifact it summarizes. This paper supplies the *why* (Scott plus the human-factors argument of §5) and the *how* (the rules below). Digest-with-zoom is the unifying statement of all three.

3.1 What to flatten, what to preserve

Scott's own distinction is the design boundary. The state may safely standardize *administrative facts it itself defines*, and must *never* standardize away the *local knowledge it depends on.*

- **Flatten (safe — the operator's own structured field):** identifiers, **claim** rows, session phases, port numbers, **commitment** status (a *commitment* being a violable obligation bound to a named actor, watched by a breach monitor), enumerated states. Administrative legibility is cheap and lossless here.
- **Preserve verbatim, never paraphrase (the agents' mêtis / the ground truth):** the diff, the test output, the error message, the reasoning trace, the exact command run. Paraphrasing these is the over-flattening.

The slogan: *summarize the STATE, link to the WORK.*

Pitfall. "Zoom to the artifact" does not fully preserve mêtis, and a political theorist will say so (§9). Mêtis in Scott is not "the detailed data underneath the summary" — it is practitioner knowledge that *resists codification entirely*: the operator's gut sense that "this agent is about to do something dumb," the local knowledge of which corners of the codebase are cursed. A read-path to the diff is more legibility, not mêtis. The honest admission — which couples Scott to Bainbridge (§5) — is that *some of the most valuable operator knowledge is exactly what the system cannot make legible, and the more the supervising automation takes over, the faster that mêtis atrophies.* What organizational-knowledge theory calls the problem of converting **tacit knowledge** into explicit, transferable form — "we know more than we can tell" [63] — is the thirty-year literature this pitfall leaves open rather than solves. The skill-retention tax (§5.4) is the design's only defense, and it is frankly the hardest part of the design to engineer.

3.2 The zoom target must be verifiable, not self-reported

There is a subtle trap inside "zoom to the real thing." If the operator zooms to the *agent's chain-of-thought* and treats that as ground truth, the map has merely moved down one level — because the reasoning trace may be fiction. **Turpin et al. 2023**, *Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting* (NeurIPS) [15] — *LLM-generated reasoning traces can be systematically unfaithful: plausible, coherent, and not actually what drove the model's output (a model swayed by reordering of options will rationalize a post-hoc reason and never mention the bias).* Therefore:

Property 3.1 (Verifiable-zoom)**.** The canonical zoom target is an artifact the operator can check *independently of the agent* — the diff on disk, the test result, the committed file. The agent's self-narration is a *secondary* lens, always labeled as such. Two LLMs agreeing (an agent acting and an LLM summarizing it) is not two checks; it may be one error reported twice.

This is why the source of truth must be the durable substrate — the coordination database and the version-control working tree — and not a model's summary of them. The digest is computed *over* artifacts the operator owns. It is also why an *adversarial* review organ (a reviewer with no stake) is structurally necessary, not optional: a swarm that reviews its own work is two LLMs agreeing, which by Prop. 3.1 is one error twice. We return to adversarial review as the authority's quality organ in §4.

3.3 The four design questions (a decision procedure)

For any read-surface — Attention Queue, briefing, resurrection digest, TUI pane — ask, in order (Table 5):

1. **Will the operator act irreversibly on this?** If yes (merge/deploy/delete), the artifact must be one keystroke away and a zoom *forced*. Never let an irreversible act rest on a digest alone (§5).
2. **What is administrative vs. mêtis?** Flatten the former; link the latter verbatim (§3.1).
3. **Who authored the summary, and can it lie?** A deterministic projection of structured state is trustworthy; an LLM summarizing another LLM is not — point its zoom at the artifact (§3.2).
4. **Is the authority itself legible here?** Every coordinator action carries a named reason (Prop. 2.1).

Table 5: A read-surface earns release through four independent proofs, in order. The rail is conjunctive: one missing proof holds the surface. [internal]

	Question	What release requires	What the reader can check
1	is the action irreversible?	the artifact is one key away	the artifact link resolves
2	is this state or mêtis?	administrative state is flattened; the verbatim trace is linked, not summarized	the trace link resolves
3	could the claim lie?	the claim resolves beyond the author's self-report	the independent evidence resolves
4	is the authority legible?	every coordinator act returns a named reason	the reason field is present and specific

Protocol 3.1. Render a digest line under the one law

Given a candidate summary line σ over a unit of swarm work:

1. **Project state.** Compute σ as a deterministic projection of structured state (claim rows, phases, enums). Never let a language model paraphrase the line itself.
2. **Attach zoom.** For every claim in σ, attach ζ: a link to an operator-verifiable artifact (the diff, the test log verbatim, the committed file). If any claim has no such artifact, the line is a Potemkin digest — refuse to ship it.
3. **Forced-zoom flag.** If the operator may act irreversibly on σ (merge, deploy, delete), set $\sigma.\mathsf{force} = \mathsf{true}$: the artifact is one keystroke away and the act is gated on a zoom (§5 sets the threshold).
4. **Reason on authority.** If σ records a coordinator act (a denial, an escalation), attach its named reason. An act with no reason is illegible authority; refuse to ship it.

Scene. Mara, 3:32 p.m. *She opens the digest for the five-PR pile. Each row reads "API-2 · charge handler · merged · 1 critique round" — state only, six words. The "fix tests" PR's row is flagged* force-zoom*, because merging it is irreversible; she presses one key and the 600-line diff and the verbatim test log open side by side. The diff deletes a migration test. The digest did not paraphrase that away — it linked her straight to it, and the lie is caught in ten seconds instead of slipping through. This is the one law doing its job.*

Exercises 4.9–4.12, page 44.

4 The Leviathan rules, it does not just watch (the authority half)

The title is *Legibility & Authority.* A read-only legibility layer is a glorified dashboard; the whole point of Hobbes is that the sovereign does not just *see* — it *rules.* This section specifies the authority half: the mechanisms by which the consented daemon actually decides, blocks, lands, and gates. Table 6 lists the organs; the engineering status of each is in Appendix A.

Table 6: Six authority mechanisms as distinct instruments crossing one consent boundary. Their inputs and outcomes differ; what they share is the recorded grant g: scope, named reason, audit identity, override, and an addressable return artifact. The engineering status of each is in Appendix A. [internal]

Mechanism	Input	Outcome	What the grant records
roadmap	evidence	a priority	the evidence weighed and the ordering it produced
review	rival claims	a gate	which claim was admitted and why the others were not
completion	a predicate	done, or not	the predicate and the artifact that satisfied it
landing	conflicts	an order	the conflict set and the serialization chosen
guard	a proposed act	a block, or passage	the act, the rule that fired and the named reason
escalation	an interrupt	an operator decision	the question raised and the decision returned

4.1 The roadmap as a versioned statement of intent

A coordinator that prevents collisions answers "what is the swarm doing?" It does not answer "is the swarm doing the *right* thing?" That question needs a single legible artifact the authority is measured against: a **roadmap** — a phase-keyed, versioned, amendable statement of intent, priority, and dependency, against which the swarm's actual output can be compared. It is not truth about the world; calling it that would turn every lesson the work teaches into drift. The authority's legitimacy is measured by *whether the roadmap stays honest*: every unit of work keys to a plan item, amendments are recorded as amendments, and drift between plan and reality is itself a legible, surfaced anomaly rather than a fact to be hidden.

4.2 Adversarial review: the quality organ

By Prop. 3.1, a swarm reviewing its own work is two language models agreeing — one error twice. The quality organ must therefore be *adversarial and conflict-free*: a reviewer with no stake in the work, running a bounded critique–refine protocol. This is the wedge's local analog of the *neutral evaluators* — the independent rating agencies — that *From Spawn to Person* and *The Harbor Economy* federate across operators: the same conflict-free-judge discipline, scoped first to one operator and then opened to a market.

4.3 Sole-responsibility roles and the specialization boundary

The single-writer discipline (§1.2) generalizes naturally from data files to system functions. When an invariant is critical, the authority assigns a **sole-responsibility role** (an avatar holding exclusive write capability over an invariant):

- *Roadmap Owner:* The single avatar authorized to commit mutations to the phase plan.
- *Test Suite Curator:* The sole agent authorized to land modifications to regression suites.
- *Release Avatar:* The sole gatekeeper authorized to execute deployment tags.

Assigning a role that way is a trade, and the trade has a number. Think of a bank: one expert teller (the sole specialist) working a single line in strict order, versus several ordinary tellers (the pooled generalists) sharing one line. The expert is faster per request but has no colleagues to absorb a burst; the pool is slower per request but almost never leaves the line standing idle behind a long job. *$M/M/1/M/M/c$ — the standard queueing-theory shorthand (**Kendall 1953** [47]) for a single-server versus c-server queue with Poisson arrivals, exponential service times, and one waiting line served first-come-first-served* — is the textbook way to price the trade.

Fix the **queueing-specialization setup**: requests for an invariant-critical function F arrive as a Poisson stream at rate λ_F; a sole specialist serves them as an $M/M/1$ queue at rate μ_{spec}, the pooled alternative as an $M/M/c$ queue of generalists at rate μ_{pool} each with utilization $\rho = \lambda_F/(c\,\mu_{\text{pool}}) < 1$ and first-come-first-served exponential service; and the objective is *mean* cost — mean response time, plus an accountability value A per unit time of demand for single-writer attribution, against a waiting cost w per request-hour. Let $C(c, \rho)$ be the **Erlang-C** delay probability (**Erlang 1917** [48] — *the probability that an arriving request finds all c servers busy and must wait, the quantity that makes a pool a pool*).

Theorem 4.1 Queueing Specialization Boundary. Under the queueing-specialization setup, sole ownership weakly dominates the pool if and only if the skill premium clears

$$\frac{\mu_{\text{spec}}}{\mu_{\text{pool}}} \;\geq\; g_A(\rho, c) \;=\; c\rho \;+\; \frac{1}{1 + \dfrac{C(c,\rho)}{c(1-\rho)} + \dfrac{A\,\mu_{\text{pool}}}{w\,\lambda_F}}. \tag{1}$$

---

verified — *corrected*. An earlier draft of this paper proposed the threshold $\tilde{g} = 1 + (c-1)\lambda_F/(c\mu_{\text{pool}} - \lambda_F) = 1 + (c-1)\rho/(1-\rho)$, on the plausible-looking grounds that it is what a naive $M/M/1$-versus-$M/M/c$ response-time comparison appears to give. A sweep against simulation *falsified* it in both directions before the belief hardened, and Eq. 1 is the replacement — the companion formal paper's Theorem 2 [55], independently re-derived and checked numerically over two thousand (ρ, c) points. We leave the wrong form on the page rather than quietly swapping it, because a paper that argues for falsification-first oversight should be legible about its own retractions: the two thresholds cross exactly once, so the superseded form was not merely conservative — it was wrong in *both* directions, and Figure 3 shades the two error regions.

At $A = 0$ — pure response time, accountability priced at nothing — the boundary reduces to

$$g(\rho, c) \;=\; c\rho \;+\; \frac{c(1-\rho)}{c(1-\rho) + C(c,\rho)}. \tag{2}$$

Three properties fix its shape: $g(\rho, 1) = 1$ (a specialist against a single generalist needs no premium at all), $g \to c$ as $\rho \to 1$ (the boundary *saturates* at the capacity ratio; it never diverges), and for two generalists it is the closed form $g(\rho, 2) = 1 + 2\rho - \rho^2$. Below the boundary, sole ownership buys strict single-writer accountability at an explicit latency price that must be budgeted rather than obscured; Eq. 1 is where that price is written down.

Numbers by hand. The result is two lines of algebra on textbook formulas, which is the right way to first believe it. Sole ownership pays iff $w\lambda_F(W_{\text{solo}} - W_{\text{pool}}) \leq A$, i.e. $1/(\mu_{\text{spec}} - \lambda_F) \leq W_{\text{pool}} + A/(w\lambda_F)$; invert, divide by μ_{pool}, and substitute the two textbook facts $\lambda_F/\mu_{\text{pool}} = c\rho$ and $\mu_{\text{pool}} W_{\text{pool}} = 1 + C(c,\rho)/(c(1-\rho))$, and Eq. 1 falls out. For $c = 2$ the Erlang-C value is $C(2, \rho) = 2\rho^2/(1+\rho)$, so the second term of Eq. 2 is $(1-\rho^2)/((1-\rho^2) + \rho^2) = 1 - \rho^2$ and $g(\rho, 2) = 1 + 2\rho - \rho^2$, checkable in three lines with no queueing theory at all.

Now the case the roadmap owner actually faces. Roadmap mutations arrive at $\lambda_F = 2/\text{hr}$; the sole owner clears them at $\mu_{\text{spec}} = 3/\text{hr}$; the alternative is three generalists at $\mu_{\text{pool}} = 1.2/\text{hr}$ each. Then $\rho = 2/3.6 = 0.556$ and the skill premium is $r = 3/1.2 = 2.5$. Erlang-C at $(c, \rho) = (3, 0.556)$ is 0.300, so $g = 3(0.556) + 3(0.444)/(3(0.444) + 0.300) = 1.667 + 0.816 = 2.483$. The premium 2.5 clears it — barely — so the sole owner wins: $W_{\text{solo}} = 1/(3-2) = 1.000$ hr against $W_{\text{pool}} = (1 + 0.300/1.333)/1.2 = 1.021$ hr, a margin of about 75 seconds per request. The superseded threshold would have returned $\tilde{g} = 1 + 2(2)/(3.6 - 2) = 3.5$ and sent the work to the pool. That is the right-hand error lens of Figure 3, met in the wild on the first realistic instance anyone tries.

Now you try: at the same three-agent pool, would a specialist at $\mu_{\text{spec}} = 2.8$/hr still be worth it on response time alone? (The premium is $2.8/1.2 = 2.33 < 2.483$ — no, and $W_{\text{solo}} = 1.25$ hr confirms it. The boundary is genuinely tight here; note that a positive accountability value A lowers g_A below g and can buy the sole owner back.)

Numbers by hand — Where the exact and superseded thresholds cross. At the round number $\rho = 0.5$: $g(0.5, 2) = 1 + 2(0.5) - 0.5^2 = 1.75$ [verified, closed form].[a] The superseded $\tilde{g}(\rho, 2) = 1 + \rho/(1 - \rho)$ gives $\tilde{g}(0.5, 2) = 2.00$ at the same point — already a visible gap. Setting $g = \tilde{g}$ and clearing denominators leaves $\rho(\rho^2 - 3\rho + 1) = 0$, whose root in $(0, 1)$ is $\rho = (3 - \sqrt{5})/2 \approx 0.382$ [verified, closed form]: below it $\tilde{g}$ over-certifies specialization (false-certify), above it $\tilde{g}$ over-rejects (false-reject), which is exactly why the superseded form was wrong in *both* directions rather than merely conservative. *Now you try:* run the falsification sweep behind this crossing (Exercise 4.18).

[a] Worked numbers carry a tag throughout: [verified] when a named script in the repository regenerates them, [internal] when only the chapter's own derivation does.

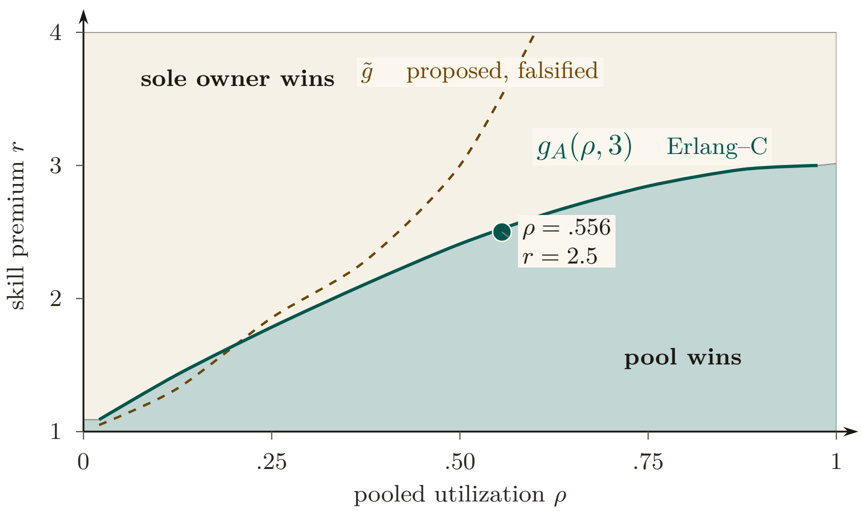

Figure 3: The specialization boundary at a three-agent pool. Below the Erlang–C boundary $g_A(\rho, 3)$ (teal), pooling wins; above it, sole ownership wins — the two regions are now filled and edged, not hatched. The superseded threshold $\tilde{g}$ (dashed) is kept for comparison: it diverges rather than saturating at the pool capacity ratio. The worked instance ($\rho = .556$, $r = 2.5$) narrowly clears the correct boundary into the sole-owner region [internal, closed form].

Pitfall. Eq. 1 prices *mean* response time under first-come-first-served exponential service. It does not price the tail, and it does not price the corner the whole sole-responsibility discipline exists for: the sole specialist *failing*, which is an $M/M/1$ queue with breakdowns and a succession protocol, not the queue modeled here. Nor does it price the thing an operator actually values in a roadmap owner — consistency of judgment across requests — which does not appear in a response-time objective at all. The boundary is a floor on the argument, not the whole argument.

4.4 Completionist obligation: completion as a verifier, not a vibe

The operator's standing complaint is the agent that *declares completion while the work is hollow* — "fixed the test by deleting it." The completionist obligation makes this structurally impossible to do silently.

Definition 4.1 (Completionist completion)**.** A **completion claim** is a commitment whose satisfaction predicate is a *verifier* derived from the task's acceptance criteria. The authority refuses to accept the claim against an unmet predicate. An *unverifiable* predicate must **fail loudly** — surface "I cannot verify this completion" — and never silently pass. This is the authority-side analog of discovery's *refuse-to-route* (§7.3): "I don't know" must beat a confident wrong answer.

The verifier is grounded in design-by-contract (**Meyer 1992**, *Applying Design by Contract* [16] — *pre/postconditions as machine-checkable obligations; the completion predicate is a postcondition*). This is the single highest-leverage new mechanism in the wedge: it turns "done" from a declaration into a checkable obligation, which is the only structural defense against the transparently-hollow deliverable.

Pitfall. Verification has a *cost gradient*, not a binary. A test going green is cheap to verify; "the architecture is right" is expensive, and reading a 600-line diff to confirm an acceptance criterion is itself attention-seconds that may exceed the agent's narration cost by an order of magnitude (the *verification asymmetry*). Def. 4.1's fail-loud is the honest floor for the *unverifiable*; §5 supplies the cost model for the *expensive-but-possible* middle.

4.5 Thoughtful landing and the regimentation/enforcement split

The authority decides *what lands and in what order*, conflict-aware. And it has teeth: the footgun guards. Here we must be precise about what a guard can and cannot do, because the difference is the difference between a safety claim that holds and one that is wishful.

Property 4.1 (Prevention vs. detection — the reserved-phrase rule)**.** A coordination state is **physically unreachable** — prevented — *only* when a constraint forbids it *inline*, before the act commits: a database uniqueness constraint that makes a double-claim impossible, or a fail-closed startup gate that refuses to run against the wrong target. A monitor that *subscribes to the log of actions already taken* is something weaker and must be named formally: **detect-and-compensate**. It observes a violation *after* the fact and then halts or repairs within a bounded window. By the reference-monitor result below, such a downstream monitor enforces no safety property at all. The phrase "physically unreachable" is therefore reserved for the constraint-enforced set — nowhere else.

Fred B. Schneider (*Enforceable Security Policies*, 2000 [14] — *an execution monitor that observes a system can enforce exactly the safety properties; a monitor placed downstream of the act it would forbid can detect a violation but cannot prevent it*) is decisive: a monitor downstream of commit enforces no safety property, only detection plus compensation within a bounded window. A post-commit anomaly detector — however useful — is therefore detect-and-compensate, not prevention. Honesty about this distinction is essential: a system that advertises detection as prevention has lied about its own safety envelope, which is the very failure (the confident-wrong report) this paper is built to eliminate.

4.6 Human-in-the-loop escalation as a typed, prioritized interrupt

The one thing that needs a human is an *escalation* (a distress signal) routed to the operator on a reserved, highest-priority lane. The mechanism wants the critical signal to win the operator's attention — but *when* the daemon stops and asks versus proceeds is a control problem (§5), and the *escalation predicate* is not an afterthought: it is a rationalized alarm philosophy (**ANSI/ISA-18.2**, *Management of Alarm Systems* [30] — *the process-control standard for alarm rationalization: every alarm has a defined priority, set-point, and operator response, with*

explicit flood-suppression). An uncontrolled-false-alarm distress lane self-destructs into habituated uselessness within days, exactly as clinical alarms do (**Cvach 2012**, *Monitor Alarm Fatigue* [31] — *85–99% of clinical alarms are non-actionable; staff learn to ignore channels that cry wolf*). We make escalation a *costly signal* (§7.3): an escalation the operator dismisses debits the agent's standing, so crying wolf has a price.
verified. The costly-signal lane has a unique equilibrium, and the debit that makes it work is a fitted quantity with a feasible range.

Theorem 4.2 The escalation threshold and its tuning band. Let an agent holding validation signal $u \in [0,1]$ earn $b(u)$ from an escalation that is upheld and pay the dismissal debit δw when it is not, with $V(u)$ the probability the operator upholds it, V monotone. The payoff $\Pi_\delta(u) = b(u) - \delta w\,(1 - V(u))$ is strictly increasing in u, so agents separate at the unique $u^\star(\delta)$ solving $b(u^\star) = \delta w\,(1 - V(u^\star))$, escalating iff $u \geq u^\star$. The set of debits under which the lane meets both an alarm-load wall and a miss-loss wall is an interval $[\delta_{\min}, \delta_{\max}]$, and it can be empty.

Proof. b is nondecreasing and $1 - V$ is nonincreasing in u, so Π_δ is strictly increasing and crosses zero at most once; a crossing exists whenever $\Pi_\delta(0) < 0 < \Pi_\delta(1)$, which the debit guarantees for $b(0) < \delta w$. The alarm-load wall bounds the escalation mass from above and is monotone in δ in one direction; the miss-loss wall bounds the suppressed mass and is monotone in the other; the feasible set is the intersection of two half-lines, an interval, possibly empty. □

This is the Spence mold, job-market signaling's separating equilibrium transplanted to an escalation lane [56]: the dismissal debit is money only a genuine signal is worth risking. The free parameter δ is fit from measured validation rates (`instrumentation.md`), and an empty band is a finding, not a tuning problem: a tight attention budget with high miss costs admits no good debit, and the fix is capacity or evidence quality. Monotone V is required throughout; a non-monotone validator breaks the separating threshold (item R14 of the research ledger, `b7_escalation_band.py`).

Numbers by hand — The threshold and the tuning band, by hand. At $\delta = 0.05, w = 1$: $u^\star = \delta w/(1 + \delta w) = 0.05/1.05 = 0.0476$ [verified, closed form] — the fraction of the validation signal u above which escalating clears its cost, one of twelve (δ, w) points the script confirms against bisection to machine precision. Under the R3 constants ($C_{\text{miss}} = 100, c_{\text{att}} = 1, C_{\text{fa}} = 5$, so $u_{\text{crit}} = 6/105 \approx 0.057$), the feasible debit interval is $[\delta_{\min}, \delta_{\max}] = [0.0350, 0.0720]$ at attention budget $A = 3.3$, and it is *empty* at $A = 2.0$ (there $\delta_{\min} = 0.396$ exceeds $\delta_{\max} = 0.072$, so no debit satisfies both walls at once) [verified closed forms, grid-confirmed]. An empty band is not a bug to tune away: it is the honest signal that the fix is capacity or evidence quality, not a better δ. *Now you try:* run the script and confirm both bands (Exercise 4.19).

Exercises 4.13–4.19, page 45.

Recall.

1. Without looking: what is the exact condition under which sole ownership weakly dominates the pool, and what does Eq. 1 reduce to when $A = 0$?
2. Name the two limiting properties that pin down $g(\rho, c)$'s shape as $\rho \to 0$ and $\rho \to 1$.
3. Why does the superseded threshold $\tilde{g}$ fail in *both* directions rather than merely being conservative?

5 The operator as a modeled entity

It is tempting to model only the *swarm* as the thing made legible, and to leave the *operator* unmodeled. But the legibility layer exists *for* the human operator, and the binding constraint at the top of the system is the operator's finite attention and fallible trust calibration. A complete design therefore treats operator attention as a resource with its own economics, parallel to the token economics of the swarm (§8).

Key idea. In a world where token cost trends toward zero, the human supervisor is the only cost that stays fixed — so the entire economic logic of the stack eventually routes through one objective: minimize *operator-attention per correct decision.* The operator is the system's reliability bottleneck, a single server with a non-elastic service rate. The 51st agent does not get a 51st operator.

The cognitive ceiling is real: **Miller 1956**, *The Magical Number Seven* [18] — *working memory holds roughly* 7 ± 2 *chunks*, which makes read-poverty (§6) a human constraint, not merely an engineering one — and **Wickens**' *Multiple Resource Theory* ([20] — *attention is not a single scalar pool but a vector over stage* $\times$ *modality* $\times$ *code* $\times$ *visual-channel; you cannot trade audio-channel attention for visual-channel attention* 1:1). The vectored model matters: the right objective is not "minimize total seconds" but *minimize peak load on the bottleneck channel*, and the ambient-vs-dense split (§7.5) is the cross-modal load-balancing mechanism that does it.

5.1 The attention objective, spined on Signal Detection Theory

A naive objective — "attention-seconds-per-correct-decision" — is Goodhart-bait: it rewards lowering the decision criterion to wave things through, because zooming costs seconds and most diffs are fine. The correct frame is **Signal Detection Theory** (**Green & Swets 1966** [17] — *an operator discriminating signal from noise trades sensitivity* d' *against a decision criterion* β*; performance is a joint function of both and of the asymmetric costs of the two error types*). The operator is trading *misses* (rubber-stamping a destructive-on-stale-premise merge) against *false alarms* (zooming on a safe diff), and the cost of a miss is wildly asymmetric to the cost of a false alarm.

Definition 5.1 (The SDT-spined attention objective)**.** Let an operator decision over an action have sensitivity d' and criterion β. With miss probability $P_{\text{miss}}(\beta)$, false-alarm probability $P_{\text{fa}}(\beta)$, miss cost C_{miss} (large: irreversible damage), false-alarm cost C_{fa} (small: wasted attention-seconds), base rate π of genuinely dangerous actions, and per-decision attention cost c_{att}, the operator objective is

$$\min_{\beta} \; \Big[\pi \, C_{\text{miss}} \, P_{\text{miss}}(\beta) \;+\; (1-\pi) \, C_{\text{fa}} \, P_{\text{fa}}(\beta) \;+\; c_{\text{att}} \, \mathbb{E}[\text{zooms}(\beta)] \Big]. \tag{3}$$

The forced-zoom policy $p(r, s, v)$ is calibrated against this loss after specifying the score distributions that determine P_{miss} and P_{fa}. Equation 3 is a decision objective, not by itself a fitted detector.

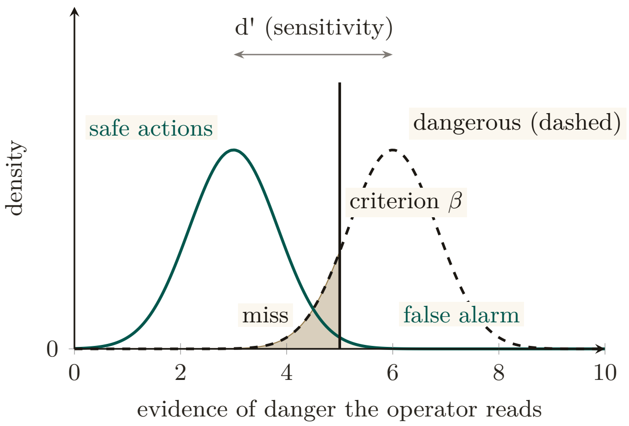

Figure 4: The operator's forced-zoom decision as Signal Detection Theory (Def. 5.1). Two named densities over the evidence the operator reads: safe actions (solid) and dangerous actions (dashed), separated by sensitivity d'. Zooming triggers when evidence exceeds the criterion β; moving β left trades the caution-tinted miss region for more of the neutral-tinted false-alarm region. Because a miss costs far more than a false alarm — at the chapter's running costs, 100 against 5, the inspect bar sits at $a \geq 0.057$ — the optimal criterion sits well to the left of the throughput optimum: idealized as two fixed Gaussians, not the fitted mixture the deployed detector would use [internal].

This is a normative claim (a Bayesian SDT decision-maker is the target to calibrate toward), not a descriptive one: naturalistic decision-making research argues skilled operators under time pressure pattern-match against recognized cases rather than compute a likelihood-ratio threshold [62], a live rival account of what the operator's instrument actually measures that this objective does not adjudicate.

Figure 4 lays the same decision out visually: the operator's criterion β trading misses against false alarms. The miss-cost term prevents the zero-review solution whenever misses have positive probability and cost. Monotonic response to stakes, irreversibility, and reputation is a policy constraint to test during calibration; it does not follow from the display without a model linking (r, s, v) to the score distributions. Before a reputation layer exists, r is a flat prior.

Numbers by hand. Equation 3 is the chapter's governing artifact for the human-factors reader, so it should be checkable on a napkin. Collapse it to its crudest useful form — zoom *every* item in an action class, or zoom *none* — and the two costs are πC_{miss} against $(1-\pi)C_{\text{fa}} + c_{\text{att}}$. Table 7 runs three real action classes through it at $C_{\text{fa}} = 1$ and $c_{\text{att}} = 0.2$ attention-seconds.

Action class	π	C_{miss}	**Never zoom**	**Always zoom**	**Verdict**
`tests/`-only formatting diff	0.001	5	0.005	1.199	auto-land
Refactor touching a public interface	0.03	40	1.20	1.17	zoom, barely
Migration dropping a foreign key	0.2	500	100.0	1.00	forced zoom

Table 7: Equation 3 on three action classes at $C_{\text{fa}} = 1$, $c_{\text{att}} = 0.2$. The first row is Mara's auto-land grant (§2.2) falling out of the arithmetic rather than being granted by taste. The third is decided by the miss term alone — $0.2 \times 500 = 100$ swamps a hundred false-alarm zooms — so the objective forces the zoom before irreversibility is separately priced. The middle row is the interesting one: at 1.20 against 1.17 the classes are within noise of each other, which is exactly the regime where a ranked queue rather than a fixed policy earns its keep.

Now you try: at what base rate does a $C_{\text{miss}}/C_{\text{fa}} = 50$ action class cross from auto-land to forced zoom? (Set $50\pi = (1 - \pi) + 0.2$, giving $\pi \approx 0.024$ — roughly one dangerous action in forty.) The boundary: this collapse assumes the operator either zooms an entire class or none of it, which is the policy Eq. 3 would produce only if P_{miss} and P_{fa} were degenerate. A fitted detector with a real criterion β does better than both columns; the table is a floor on the decision, not the decision.

5.2 Trust calibration is multi-dimensional, and the vigilance decrement is real

Showing the operator a beautiful digest does not make them appropriately reliant on it (**Lee & See 2004**, *Trust in Automation* [21] — *trust has at least three bases (performance, process, purpose) and calibration depends on* ***resolution*** *(can the operator tell where automation is reliable from where it is not?) and* ***specificity*** *(is trust differentiated across functions and time?)*). The implication is sharp: a *single* "approve-without- zoom rate" and a *single* global forced-zoom knob is mis-calibration by aggregation. An operator who correctly trusts the swarm on refactors and correctly distrusts it on migrations shows a middling *global* catch-rate, and a global knob mis-serves both. Trust calibration must be *function-specific and history-dependent* — per (action-class, agent), not one scalar.

And the canary-defect catch-rate mechanism — inject known-bad diffs at a sampled rate, raise friction when catch-rate drops — is a *vigilance task*, monitoring for rare signals, and the vigilance literature predicts it will degrade for reasons unrelated to swarm danger (**Warm, Parasuraman & Matthews 2008**, *Vigilance Requires Hard Mental Work and Is Stressful* [25] — *detection of rare signals degrades sharply within 20–35 minutes, worse at low signal probability*; the classic result is **Mackworth 1948** [26]). The honest, citable tension: defeating the vigilance decrement *costs attention-seconds* (canary rate must be high enough to beat the decrement, which fights Eq. 3). Catch-rate must therefore be *time-on-task-corrected*, and the right countermeasure is to force breaks, not just rotate zooms.

Pitfall. Campbell's Law eats the governor. The moment "operator catch-rate" becomes a metered invariant that raises friction when it drops, the operator is incentivized to game it — memorize canary patterns, develop a "looks-like-a-canary" heuristic orthogonal to real defect-finding (**Campbell 1979** [27] — *the more a quantitative indicator is used for decision-making, the more it distorts the process it monitors*). The accountability instrument corrupts the accountability it measures. We do not claim the canary governor cleanly closes the loop; it is net-positive only if canaries are drawn from a refreshed, costly-to-fake distribution and the catch-rate is one signal among several, never the sole gate.

5.3 Situation awareness needs projection, not just perception

Digest-with-zoom is overwhelmingly a perceive-and-comprehend instrument. But **Endsley 1995** (*Toward a Theory of Situation Awareness* [23] — *SA has three levels: perception, comprehension, and* ***projection*** *— what the system will do next*) puts projection at the top, and **Endsley & Kiris 1995** (*the out-of-the-loop performance problem* [24] — *operators removed from active engagement lose the dynamic SA needed to take over when automation fails;* intermediate *automation can yield* better *SA than full automation because the operator stays cognitively engaged in projection*) shows the deeper danger: a digest that summarizes only the *present* is an SA-destroying instrument no matter how good its zoom. The authority half therefore needs a first-class *projection* mode — "what is the swarm about to do, and what will the state be in ten minutes?" — validated by a freeze-probe (pause the swarm, ask the operator to predict the next three merges, score accuracy). The forensic time-axis (§12.1) is the *backward*-looking complement; projection is the missing *forward* one.

5.4 The Bainbridge–Scott convergence and the abdication gradient

Lisanne Bainbridge 1983 (*Ironies of Automation* [22] — *the more reliable an automated system becomes, the less practiced and situationally aware its human supervisor becomes, so the human is least equipped to intervene at exactly the moment intervention is needed*) and Scott are the *same warning at two scales*: the digest erases the swarm's mêtis, and the digest erases the *operator's* mêtis. As the supervising automation gains autonomy — executing longer plans end-to-end — the operator's role slides doer → approver → rubber-stamp → absent — the *abdication gradient*, stated here as a hypothesis, not a measurement. The design's primary active defense is the **Inception Canary** (a *consented, blinded* vigilance probe), moving beyond static "skill-retention taxes" (such as arbitrary manual busywork, which causes attention residue and degrades focus [28]).

Protocol 5.1. The Inception Canary protocol

To continuously measure and maintain operator situation awareness without synthetic busywork. Operators enroll knowingly: the program's existence, rate ρ_{canary}, and stakes are disclosed at onboarding; only the *which* and *when* are blinded:

1. **Blinded holdout replay.** An out-of-band Compaction Sidecar (§8.2) draws from a curated holdout of *historical, already-fixed* defects (an off-by-one boundary condition, an unindexed query, a removed regression test), re-materializes one inside a sandboxed replay of a past work item at rate ρ_{canary}, and routes it through the review surface indistinguishably from live traffic. No anomaly is ever injected into a live pull request or pending action: probes are replays with known ground truth, so a missed canary costs vigilance points, never production correctness.
2. **Attention queue surfacing.** The item is placed in the Attention Queue with standard high-regret scoring.
3. **Dual adjudication:**
   - *Rubber-stamp failure:* If the operator approves without expanding the forced-zoom diff, the daemon halts execution pre-commit, logs an Inception Canary Failed event, and tightens the operator's autonomous policy lease (lowering the threshold at which future agent actions require manual confirmation).
   - *Vigilant detection:* If the operator zooms, identifies the flaw, and rejects it, the daemon logs Canary Defeated and increases the *Operator Trust Score*, unlocking extended autonomous execution grants (e.g., unattended batch merges).

This transforms supervisory vigilance from a punitive chore into a closed verification loop that rigorously measures human situation awareness — with the operator's informed consent, on ground truth the system already knows, and with provenance every probe carries so post-hoc audit can distinguish canaries from live items in one lookup.

5.5 The operator's instrument

The operator of this chapter looks at something. In the product that something is the operator console, a GPU-native application that opens on one full-window conversation, the *Mission* view, and opens inspectors from it: fleet, cockpit, claims, parley, lineage, the cost ledger, the active harnesses, the interruptions awaiting a human, and some two dozen more, each pane fetching one route from the daemon and rendering the records that route returns. Two things about it follow from this chapter; neither flatters it, and neither is a complaint.

First, every pane is a projection, and the information floor applies to each one separately. A pane is a channel with a coverage, an omission rate, and a review latency, and the front matter's question — which channel, and what can it not show? — has a separate answer for every pane. The claims pane shows what cooperating clients registered and nothing about what a non-cooperating process wrote; the lineage pane shows the witnessed record and is blind to the unrecorded fork; the cost ledger shows a protected meter's count (and, being a meter, has no opinion about the cause). A console that draws all of them beautifully on one screen has not lowered the floor; it has made the floor easier to forget. The design rule this chapter derives, that every digest line must carry a route back to its artifact, is the rule the panes are held to, and it is what makes the thing an instrument rather than a dashboard.

Second, the console is not only a viewer. It issues commands: claim and release, begin and done, a note, a lane message, a dispatch rejection, an interruption of an agent, a kill, a steer. Each of those passes through the daemon and lands in the same single-writer record the agents write to, so the console is an authority surface in the COORDINATED mode, with the same ceiling as any other cooperating client, and its commands are the operator's own work units, receipted like anyone else's. The rendering stack behind the window (a native shell with vector surfaces for the bespoke charts) is a separate decision record and carries no claim in this book. What carries a claim is the grade: the default build and the continuous integration gate compile and test the console's model on every platform, while the window itself builds on one platform only, behind a feature flag, in one dedicated job. The grade is *built, weakly*, and the weakness is a specific one — the window most readers will picture when they hear "console" is the part of it the tests exercise least.

5.6 Modes of cooperation

The chapter has spoken of one operator and many agents. The product's own design binder names five ways people and agents actually share a repository, and each is a supervision regime in this chapter's sense: it fixes who holds write authority, what the person must be able to see, and how a conflict is settled. The binder's names for the two kinds of agent are worth keeping: *Voyagers* do the work, and *Longshoremen* watch it, keeping the continuous-integration state, the memory, the claims, and the transcripts in view.

Solo with staff One person writes. Longshoremen observe and suggest; they hold no write authority. The person must see the staff's suggestions as suggestions, and the legibility cost is the smallest of the five: nothing changes that the person did not type.

Pair with agent One person and one Voyager. The agent may suggest, patch, test, and respond to guidance; the person directs. Authority is shared turn by turn, and the person must see every patch before it lands, which is the regime the forced-zoom diff of the canary protocol above was designed for.

Ensemble Several Voyagers, partitioned by context, with a Longshoreman tracking the dependencies between partitions and the parleys that resolve them. Write authority is distributed by the partition, so the cost of seeing is paid per partition and the tracking Longshoreman is the digest this chapter prices.

Nightshift Agents work unattended under strict budgets, tests, and pull-request rules, then present artifacts and transcripts for review. The agent writes within its budget; the person sees the artifacts, not the acts. The binder's rule is the one to remember: a Nightshift agent may open a pull request under budget, but merging requires explicit rules and green gates. This is the regime with the largest gap between what happened and what the person saw, and therefore the one the information floor bites hardest.

Harbor co-edit People and agents as co-equal peers in one governed editor buffer, with claims, semantic conflicts, and salvage. Write authority is co-equal and conflicts are settled by the claim machinery of *The Single-Writer Kernel*: a same-symbol write is blocked or forced into a parley, a dependency conflict warns, a low-confidence overlap is advisory, and a rename or a delete draws the stronger review.

Three governance rules cut across the modes and are the chapter's normative claim made operational: the operator controls the harbor's policy while the agents' souls control their commitments within it; a Longshoreman must not silently perform a consequential action outside its mandate; and any action that requires a human in the loop must be shown as such in the instrument, not inferred from its absence. Read as regimes, the five modes are five points on this chapter's boundary diagram, ordered by how much of the write authority has left the person and, in the same order, by how many bits of digest the person must read to remain the operator rather than the audience. The modes are a design invariant; the costs per mode are an empirical hypothesis, and the study named in the front matter is where they will be measured.

Exercises 4.20–4.24, page 45.

Recall.

1. Without looking: what three quantities multiply to give the derived regret head, and what does calibration require of the anomaly term for that product to equal expected unrecoverable loss?
2. State in one sentence why a global trust knob cannot be the right calibration unit.
3. What is Campbell's Law's specific failure mode for a canary program, and the one governor that survives it?

6 Read-poverty: the binding constraint at scale

A single operator can hold two coding agents in their head. They cannot hold two hundred. Spin up two agents and coordination is free: you read both their notes, eyeball both their claims, route a question by recognition. Spin up fifty and an agent that wants to ask "who owns the OAuth refactor?" has no move except to dump the session roster and scroll. It guesses, cold-DMs the wrong agent, and re-derives from scratch a design decision another agent settled an hour ago and wrote into a note nobody will ever read again.

The swarm has not run out of information. It has drowned in it. Every agent emits sessions, claims, notes, diffs, skill tags, and — eventually — reputations, far faster than any single participant can read them.

Definition 6.1 (Read-poverty)**. Read-poverty** is a regime in which legible state accumulates faster than it can be consulted, so participants fall back to $O(n)$ eyeball search over the population, or, worse, act blind.

Thesis. Read-poverty, not write-contention, is the binding constraint on swarm scale. The write side has known fixes — claims, locks, merge queues, and anomaly detection — which mature coordination systems already provide. The read side has been left implicit, and it is where the next order of magnitude is won or lost.

The cure is the move every database made when table scans stopped scaling: build an **index**. A *directory* is to agents what an index is to rows — it pays a write-time cost (maintaining the structure) to buy read-time speed (answer a query in $O(\text{query})$ instead of scanning $O(n)$ participants). Read-poverty is also a human constraint (Miller's 7 ± 2, §5) and a legibility argument in Scott's sense: a directory is a state-like act of legibility over the swarm, and a directory entry over-flattened into a verdict that hides what mattered ("this agent owns auth," omitting that its last three auth pull requests were reverted) is the failure. Every directory entry must therefore be a lens that zooms (Def. 3.1).

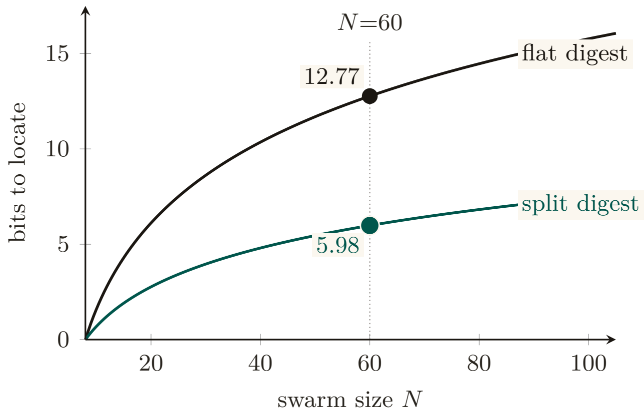

Figure 5: Bits an operator must read to locate the critical artifact, at the lower bound of Thm. 6.1 with m=8, k=2. One flat digest serving both readers jointly needs Floor$(N, 4, 8)$ bits; two split digests, one per reader, each need only Floor$(N, 2, 8)$. At N=60 the split digest costs 5.98 bits against the flat digest's 12.77 — a 2.13× penalty, not the 2× that halving the readers would predict. The committed sweep CSV (`r1-floor.csv`) is not yet present, so the two curves are the chapter's own closed-form Floor formula rather than simulated points; only the N=60 values it works by hand are marked [internal].

6.1 The value curve across n (the wedge must work at n=1)

Read-poverty is a *scale* disease, but the wedge ships to a solo developer who often runs one or two agents. The legibility layer must be valuable at n=1 — the briefing, the attestation, the footgun guard, the honest completion gate all pay off immediately — and *gracefully grow* into the read-poverty regime (Figure 5, right panel). Discovery is free at n=2, the bottleneck at n=50, the whole system at n=500. This value curve *is* the three-layer cure of the next section: existence (the registry), relevance (ranked routing), and trust (reputation, deferred to the accompanying chapters). The layers are strictly ordered: relevance over a stale registry returns confident garbage; reputation over an unranked registry has nothing to attach a score to.

6.2 A legibility lower bound

Section 3 forbade the Potemkin digest by fiat. We can also state the prohibition as a theorem: there is an information floor below which a digest *cannot* be a faithful zoomable index, no matter how it is written.

Theorem 6.1 Legibility lower bound with a review budget. Let N artifacts contain an unknown critical subset S of size k. A digest maps each possible S to a review set $\hat{S}$ with $S \subseteq \hat{S}$ and $|\hat{S}| \leq m$, where $k \leq m \leq N$. Any such zero-miss digest needs at least

$$\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$$

bits in the worst case. Exact identification ($m = k$) recovers the $\log_2 \binom{N}{k}$ bound; allowing the operator to open everything ($m = N$) correctly reduces the bound to zero.

Proof sketch. There are $\binom{N}{k}$ possible critical subsets. One digest message whose review set has size at most m can safely cover at most $\binom{m}{k}$ of them, namely the k-subsets contained in that review set. Therefore at least $\binom{N}{k}/\binom{m}{k}$ distinct messages are required. Taking base-two logarithms gives the bound. If fewer messages are available, some critical subset is not contained in its message's review set, producing a false negative — an unopened critical artifact. That is exactly the Potemkin failure of Def. 3.1. Hence the stated log-ratio is a lower bound for a zero-miss digest with review budget m. □ □

verified.[1] The analogy that carries the proof: a digest is not a description of the night's work but a *message* that selects, from a codebook shared with the overseer, which m items to open. A B-bit message distinguishes at most 2^B codewords, and each codeword covers only $\binom{m}{k}$ of the $\binom{N}{k}$ ways the critical items could be placed. The misread to preempt: the floor does not say a digest needs $B^\star$ bits to be useful. It prices the zero-miss guarantee only; cheaper digests are fine the moment misses are tolerated, and the two-constraint rate below prices exactly that.

The falsification experiment. The theorem makes a sharp prediction: a B-bit digest's best zero-miss probability is $\min\big(1,\, 2^B\binom{m}{k}/\binom{N}{k}\big)$, so any scheme dipping below the implied miss curve refutes the formalization. Three encoders were run against a shared randomized-cover codebook: an *oracle* that sees the true set and addresses the best codeword its B bits can reach, a *noisy* encoder (feature signal-to-noise ratio near 1), and a *random* one, with 4,000 placements per budget point, sweeping B through the floor at $N{=}60$, $k{=}2$, $m{=}8$. Result: 0/16 violations; the oracle hugs the floor from above and the others pay a premium [internal, `a7_experiment.py`].

A wrong turn, reported. A first implementation handed the oracle a perfect per-item score vector and charged B bits only for resolving rank ties: 8/14 apparent violations. That was not a refutation but an unpriced side channel. The identity of the critical set leaked outside the metered message. Rebuilding the model so that B is a literal message length gating an encoder and decoder pair restored agreement. The transferable rule: a simulation that appears to beat an information bound is leaking through an unpriced channel; audit the meter before doubting the theorem, and before trusting the refutation.

Numbers by hand. Take $N = 1000$ artifacts with $k = 3$ critical among them and a review budget of $m = 10$ opens. The floor is $\log_2\binom{1000}{3} - \log_2\binom{10}{3} = 27.31 - 6.91 = 20.4$ bits — about two and a half bytes the digest cannot avoid spending, no matter how well written, just to point at the right three files. Widen the budget to $m = 100$ and the second term grows to $\log_2\binom{100}{3} = 17.3$, cutting the floor to 10.0 bits: buying the operator more opens is exactly buying the digest fewer bits, at a rate the formula prices. *Now you try:* at $N = 60$, $k = 2$, $m = 8$ — the configuration the companion paper's sweep hammers hardest — what is the floor? $(10.79 - 4.81 = 5.98$ bits; the sweep found no encoder beating it in 16 tries.)

The bound is the formal reason *forgetting must be selective, not lossy-by-volume*: a digest may drop inert detail freely, but it must retain enough to *point at* every critical artifact, which is precisely the "summarize the state, link to the work" rule. It also bounds recursive summarization (§8.4): each re-summarization that drops below the floor on the critical set is irreversible information loss, which is why one must compact from the durable artifacts, never from the previous summary.

The other half of the price: what a zoom costs. The floor above prices only the first stage, the digest that narrows N artifacts to a flagged set of size F. It says nothing about the second stage, which is where Def. 3.1 sends the operator. Two results complete the account. The first prices the digest once misses are tolerated; the second prices the zoom.

Model each artifact's critical status as i.i.d. $X \sim \mathrm{Bern}(p)$, let the digest emit a per-item flag $\hat{X} \in \{0, 1\}$, and price the two failure modes separately:

$$R(\delta, f) \;=\; \min_{P_{\hat{X}|X}} \; I(X; \hat{X}) \quad \text{s.t.} \quad \Pr(X{=}1, \hat{X}{=}0) \le \delta, \qquad \Pr(\hat{X}{=}1) \le f.$$

[1] A standalone, submission-form version of the results folded into this subsection, §7.2 and §8.2 is *The Price of a Summary* (research paper 1) [49]; the sweeps cited here are its artifacts, regenerated at seed 20260816.

This is a custom formulation, deliberately: classical rate–distortion charges one scalar distortion, which cannot express that oversight prices a missed critical differently from a wasted open. The nearest published lines are expansion coding with one-sided error for continuous sources [53] and the rate–distortion–classification theory for Bernoulli sources [54]; neither prices false negatives as a separate constraint for binary sources.

Theorem 6.2 The two-constraint digest rate, pinned. For $X \sim \mathrm{Bern}(p)$ with $0 \le \delta < p$ and $p - \delta \le f < 1 - \delta$, if $f < 1 - \delta/p$ (the flag rate below which an X-independent flagger cannot meet the miss constraint), both constraints bind at the optimum and the minimizing joint is fully pinned:

$$q_{11} = p - \delta, \qquad q_{10} = \delta, \qquad q_{01} = f - (p - \delta), \qquad q_{00} = 1 - f - \delta,$$

feasible iff all four are nonnegative, and $R(\delta, f) = I(X; \hat{X})$ evaluated at this joint. At the zero-miss, tightest-flag corner, $R(0, f \to p) = H(p)$: the digest must spend the source's full entropy. Past the line $f \ge 1 - \delta/p$ an independent flagger already meets the miss budget on its own and the true rate is exactly 0.

Proof. $I(X; \hat{X})$ is convex in $P_{\hat{X}|X}$ and decreases along the feasible ray toward the independent coupling, where it is 0. An independent flagger $\hat{X} \sim \mathrm{Bern}(\phi)$ has miss mass $p(1 - \phi)$, so it meets the miss constraint only when $\phi \ge 1 - \delta/p$; when $f < 1 - \delta/p$ that coupling lies outside the flag budget, so the minimum sits on the boundary where any remaining slack in either constraint could otherwise be spent moving further along the ray, and both constraints bind. Binding constraints fix $q_{10} = \delta$ and $q_{11} + q_{01} = f$; with the source margin $q_{10} + q_{11} = p$ this pins all four cells. At $(\delta, f) = (0, p)$ the joint is diagonal, $\hat{X} = X$, and $I = H(p)$. □

A conservative digest over-flags to be safe: it emits a flagged set of size F containing the k true criticals with $k \ll F$. The naive second stage opens all F. The adaptive second stage plays twenty questions with group queries: open a *group* of flagged items at once (an aggregate check cheap at group level, a combined diff scan or a batched test run), learn whether the group contains any critical, and split only the groups that do. The analogy transfers because its defining relation transfers: halving a set costs one question and eliminates half the candidates only when positives are rare in the set. The misread to preempt: twenty questions finds *one* answer in $\log_2 F$ queries; with k answers the tree forks, and the accounting below is precisely the price of the forking.

Theorem 6.3 Zoom advantage. Let a flagged set of F items contain an unknown set P of $k \ge 1$ positives, and let a group query on S report whether $S \cap P = \emptyset$. Adaptive halving (query the flagged set; discard any group that tests negative; report any positive singleton; otherwise split the group into halves of sizes $\lfloor |S|/2 \rfloor, \lceil |S|/2 \rceil$ and recurse on both) identifies P exactly, using a total number of group queries

$$Q \;\le\; 2k \left\lceil \log_2(F/k) \right\rceil + 4k \qquad \text{(and always } Q \le 2F - 1\text{)},$$

for every placement of the k positives, against F opens for flat inspection. The leading constant is tight: for F and k powers of two, evenly spread positives force $Q = 2k \log_2(F/k) + 2k - 1$. Written in the flagged-set density $d = k/F$ the guaranteed advantage is

$$\mathrm{adv}(d) \;=\; \frac{F}{2k\lceil \log_2(F/k) \rceil + 4k} \;=\; \frac{1}{d(2\lceil \log_2(1/d) \rceil + 4)},$$

which exceeds 1 exactly on $d \leq 1/12$.

Proof. Every group the algorithm queries is a node of the halving tree T of $[F]$: the root is the full flagged set, each internal node's children are its two halves, leaves are singletons. A node of size s has children of size at most $\lceil s/2 \rceil$, so every node at depth d has size at most $\lceil F/2^d \rceil$, T has depth $D \leq \lceil \log_2 F \rceil$, and T has exactly $2F-1$ nodes, which gives the unconditional bound.

Call a node *positive* if its group meets P. The algorithm splits exactly the positive non-singleton nodes it visits, and every queried node is either the root or a child of a split node; writing S^+ for the number of positive non-singleton nodes of T, $Q \leq 1 + 2S^+$. The nodes at any fixed depth are pairwise disjoint subsets of $[F]$, so each positive item lies in at most one of them, and at most $\min(k, 2^d)$ nodes at depth d are positive. Charge the positive nodes in two regimes. At the top of the tree, $d \leq \lfloor \log_2 k \rfloor$, charge each positive node to its level: at most $\sum_{d=0}^{\lfloor \log_2 k \rfloor} 2^d \leq 2k-1$ in total. Below the fan-out, $d > \lfloor \log_2 k \rfloor$, charge each positive node to one positive item it contains; by disjointness within a level each of the k positives absorbs at most one charge per level, and non-singleton nodes exist only at depths $d \leq D-1$, so the number of charging levels is $D - 1 - \lfloor \log_2 k \rfloor < (\log_2 F + 1) - 1 - (\log_2 k - 1) = \log_2(F/k) + 1$; an integer strictly below $\log_2(F/k)+1$ is at most $\lceil \log_2(F/k) \rceil$, so these levels contribute at most $k\lceil \log_2(F/k) \rceil$ charges. Summing, $S^+ \leq k\lceil \log_2(F/k) \rceil + 2k - 1$ and $Q \leq 2k\lceil \log_2(F/k) \rceil + 4k - 1$. Correctness is immediate: a discarded group is certified free of positives, every positive survives discarding at every level, and each is eventually isolated and reported.

For tightness take $F = 2^a$, $k = 2^b$, $b \leq a$, positives at positions $\{iF/k : 0 \leq i < k\}$. Every node at depth $d \leq b$ contains a positive, so all $2^{b+1} - 1 = 2k - 1$ nodes down to depth b are queried; below depth b exactly k nodes per level are positive, each is split at levels $b, \ldots, a-1$, and each split queries two children; the total is $(2k-1) + 2k(a-b) = 2k \log_2(F/k) + 2k - 1$. The density form follows by dividing F by the bound, and the step function $d(2\lceil \log_2(1/d) \rceil + 4)$ first touches 1 at $d = 1/12$. □

Positioning. None of the group-testing machinery is new mathematics: Dorfman introduced pooled testing in 1943 [50], and Hwang's generalized binary splitting [51] finds k defectives in $\log_2 \binom{F}{k} + O(k)$ tests, essentially matching the counting lower bound $\lceil \log_2 \binom{F}{k} \rceil \geq k \log_2(F/k)$ that any adaptive strategy with binary outcomes must pay; the full theory is Du and Hwang [52]. Plain halving pays a factor $2 + o(1)$ over Hwang because it re-queries both children of every split rather than inferring one subgroup and searching within the other. The theorem's contribution is the explicit worst-case constant for the exact procedure the digest architecture runs, proved tight, which turns the harness's observed $\approx 2\times$ gap over the ideal $k \log_2(F/k)$ from an empirical curiosity into the algorithm's provable constant. A dedicated check re-runs the exact procedure: at F=2500, k=10, mean 162.6 opens over 150 random placements, maximum 183, all under the bound of 200; random sweeps over $F \in [16, 5000]$, $k \in [1, 50]$ and adversarial evenly-spread and clustered placements never exceed the bound; and the tightness configuration F=4096, k=32 lands exactly on $2k \log_2(F/k) + 2k - 1 = 511$ [internal, `zoom_bound_check.py`].

The composition window. The two prices are drawn from one budget, and they compose only inside a window that can be stated exactly. Stage one sets stage two's density: by Theorem 6.2 both constraints bind, so the flagged set holds $(p-\delta)N$ real criticals among fN flags and $d = k/F = (p-\delta)/f$. Since $\mathrm{adv}(d) > 1$ exactly on $d \leq 1/12$, *the two stages compose profitably iff the flag budget is at least twelve times the residual positive rate*, $f \geq 12(p-\delta)$ [verified from the two closed forms]; the measured mean crossover is about twice as forgiving, $d \approx 0.165$ [internal, `zoom_bound_check.py`]. At p=0.05, δ=0 the window opens only at $f \geq 0.60$: the digest must flag 60% of the night's artifacts before the second stage is guaranteed to beat reading the flags. Inside the window the flags stage one adds are cheap in the second stage, because k is fixed by the source and doubling F costs only the logarithmic term, $2k$ further opens. Outside it the second-stage cost is linear in F, because reading all F flags is then the better procedure. At the dense end the advantage inverts outright: at F=100, k=90 the run costs 199 opens against 100 flat [internal], which is the unconditional $2F-1$, reached because at that density essentially every node of the halving tree is positive. The design consequence for this chapter is therefore conditional, and the condition is now a number: "summarize the state, link to the work" is cheap when the digest over-flags generously and the zoom is adaptive, and the operator's dial (bits on the digest, opens on the zoom) has a favorable exchange rate exactly in the sparse-flag corner.

Numbers by hand — The Bernoulli view of the floor, and the zoom-advantage numbers. Write the flagged-item rate as a probability p instead of a fixed pair (N, k): the zero-miss floor becomes the binary entropy $H(p) = -p\log_2 p - (1-p)\log_2(1-p)$ per artifact in the large-N limit — the same result as Thm. 6.1, in a different notation. At $p = 0.05$: $H(0.05) = 0.2161 + 0.0703 = 0.2864$ bits/symbol [verified, Cover–Thomas], the sparse-flag rate this chapter treats as typical (a digest of 105 artifacts with 6 flagged is $6/105 \approx 0.057$, close enough for the same number to apply). Tolerating a miss rate $\delta > 0$ only ever lowers the floor: the two-constraint rate $R(\delta, f)$ drops to 0.1864 bits at $f = 0.10, \delta = 0$ and to 0.0087 once $\delta = 0.04, f = 0.06$ [internal, `b1_frontier.py`] — $R(0, p) = H(p)$ is the corner, not the whole curve.

For the zoom side, the script's own $k\log_2(F/k)$ column at $F = 2500$ flagged, $k = 10$ critical reads $79.7 \approx 80$: an order-of-magnitude sanity check, tiny next to 2500 flat opens. The simulated adaptive-halving harness measures 163.7 opens at that same point, comfortably inside the 200-open guaranteed worst case quoted above, for a realized $2500/163.7 \approx 15.3\times$ reduction [internal, `b1_frontier.py`] — better than the guarantee's own $12.5\times$, exactly as a worst-case bound should allow on a typical run. *Now you try:* run the script and read the F=1000, k=20 row (Exercise 4.30).

Where this stops. Theorem 6.1 lower-bounds the zero-miss guarantee in a worst-case combinatorial model with a data-independent decoder; it says nothing against cheaper digests at nonzero miss (Theorem 6.2's job) and nothing average-case. The falsification result 0/16 tests the encoder and decoder formalization at one swept regime (N=60, k=2, m=8); the noisy-encoder curve in that experiment barely beats random at unit signal-to-noise ratio and is not evidence that noisy digests are near-optimal. The frontier $R(\delta, f)$ is an i.i.d. Bernoulli model of critical status, a modeling commitment rather than a law, and its pinned closed form holds only while $f < 1 - \delta/p$; an earlier plot that evaluated the formula past that line showed a spurious uptick near $\delta/p \to 1$, a domain-of-validity error and not a feature of the frontier. Theorem 6.3 assumes a reliable group query at unit cost; correlated or noisy group checks, and group costs growing with $|S|$, change the accounting. Plain halving is a factor of about two from Hwang's optimum by design: the theorem bounds the deployed algorithm, not the best one. The two halves compose only for $f \geq 12(p - \delta)$, so at p=0.05 they are not jointly instantiated at the flag budgets the frontier tabulates.

Exercises 4.25–4.30, page 45.

Recall.

1. Without looking: what two terms does the legibility lower bound subtract to get the bit floor, and what happens to it as the review budget m grows to N?
2. State the zoom theorem's opens bound in one sentence, and the one condition (sparse vs. dense flags) it needs to hold.
3. Why is a floor stated for an oracle-known critical set not automatically a floor against an adversary who controls what looks critical?

7 Discovery: the directory substrate and the split ranker

A performative cannot be addressed until the swarm knows who exists. The *directory substrate* — the existence layer a message must address — belongs with the discovery story, so we treat it here.

7.1 Existence: the yellow pages (a solved shape)

The canonical prior art is forty years old. **FIPA** (*Foundation for Intelligent Physical Agents*) made a **Directory Facilitator** (DF) — *the mandatory "yellow pages" agent: others register the services they offer and query to find agents that offer a needed service* (FIPA Agent Management Specification, FIPA00023 [37]) — a required component of every conformant platform, alongside the **Agent Management System** (AMS, the "white pages" for *identity* lookup). Two FIPA choices matter: the DF is *push-based* (agents self-advertise), and DFs may *federate* (the seed of cross-operator discovery). The DF has been re-invented almost beat-for-beat for LLM agents (**Ren et al. 2025**, *Agent Name Service* [38] — *a DNS-inspired naming and capability-resolution layer with public-key-backed identity*, which is FIPA's DF with a 2025 threat model).

The existence layer is the easy part: a coordination database that records the live population, their stated purpose, and their file claims is a white-pages directory, and a vector index that embeds each agent's declared capabilities and answers nearest-neighbor queries is a yellow-pages directory over skills. Existence is not the gap; the layers above it are.

7.2 Relevance: the split ranker (the headline result)

A flat registry answers "does an OAuth specialist exist?" It does not answer "of the fifty live agents, who should I ask about *this* OAuth bug, given who touched these files an hour ago and who wrote the design note?" That is a *ranking* problem. The **discovery ranker** scores each candidate c against a query q as a clamped weighted sum of normalized sub-scores:

$$\begin{aligned} z(c,q) &= w_f\,\mathrm{file}(c,q) + w_n\,\mathrm{note}(c,q) + w_i\,\mathrm{ident}(c,q) \\ &\qquad + w_s\,\mathrm{skill}(c,q) + w_e\,\mathrm{epis}(c,q) - w_\ell\,\mathrm{load}(c), \\ \mathrm{fit}(c,q) &= \mathrm{clamp}\big(z(c,q),0,1\big). \end{aligned} \tag{4}$$

Table 8 is the whole signal inventory.

Signal	What it measures	Push or pull
file	Recent claims on the files named in the query	*Pull* — the claim record, not a self-report
note	Text similarity over the candidate's written notes	*Pull* — written for another purpose
epis	Text similarity over past episodes the candidate closed	*Pull* — outcomes, already witnessed
ident	Declared role and identity of the candidate	*Push* — asserted, cheap to inflate
skill	Declared/embedded capability, nearest-neighbor in the skill index	*Push* — asserted, and stale by default
load	How busy the candidate already is (penalty, $-w_\ell$)	*Pull* — observed occupancy

Table 8: The six sub-scores of Eq. 4, sorted by whether the candidate *told* the ranker or the ranker *observed* it. Four of the six are pull signals, and that is deliberate: push signals are the ones an agent can inflate at no cost.

The design deliberately blends *push* signals (declared capability — gameable and prone to staleness) with *pull* signals (demonstrated capability: what the agent actually *did*, in which there is no self-report to lie in); Table 9 sets out the split and the reason pull signals carry more weight. One discipline is non-negotiable: **never classify free text by keyword lists.** Keyword matching has catastrophic recall, because the category's vocabulary cannot be enumerated in advance; the only exact-string match permitted is over *structured identifiers the operator controls* (file paths, enum tags). Every free-text signal uses a ranked-retrieval scorer (BM25, character-bigram TF–IDF, or learned embeddings).

Table 9: Push and pull evidence for the same capability claim. ○ is a self-declaration, ● is residue produced while doing work for another purpose, × is a declared item that does not resolve, and a dash is nothing on record. A and B make the same identity and skill claims; only B leaves the complete file, note, episode and current-load trail, so the ranker, following independently addressable evidence rather than confidence in self-report, ranks B first. [internal]

Evidence	**Kind**	**A**	**B**	**What it would take to fake**
identity	push: declared	○	○	nothing; it is asserted
skill	push: declared	○	○	nothing; it is asserted
file	pull: residue	●	●	a forged diff under a real commit
note	pull: residue	—	●	a note with a ledger address and a timestamp
episode	pull: residue	×	●	a whole prior episode with its outcomes
current load	pull: residue	—	●	the daemon's own claim table
verdict		small verified weight	ranks first	

It is tempting to claim that this discovery ranker and the operator-facing *attention queue* are *the same function* with different weights. The claim is elegant and, on inspection, false: the two readers have incompatible loss functions. We state the distinction as a theorem.

Theorem 7.1 No universal monotone identification of the two heads. On any item domain containing both high-fit/low-risk and low-fit/high-risk examples, there is no strictly increasing scalar transform that identifies discovery fit with operator regret. The two rankers may share candidate generation and recency decay, but apply distinct scoring heads. Discovery uses capability fit (Eq. 4); the attention queue uses regret-if-ignored:

$$\mathrm{regret}(x) \;=\; \underbrace{\mathrm{stakes}(x)}_{\text{magnitude}} \times \underbrace{\mathrm{irreversibility}(x)}_{v^{-1}} \times \underbrace{\mathrm{anomaly}(x)}_{\text{surprise}}. \tag{5}$$

No universal monotone transform of one scalar yields the other on that domain.

Proof sketch. Suppose, for contradiction, that there is a strictly monotone map ϕ with $\mathrm{regret}(x) = \phi(\mathrm{fit}(x))$ for all items x, so the two heads induce the same ranking. Construct two items. Item A is a highly capable, perfectly on-topic completion of a safe, fully reversible task: $\mathrm{fit}(A)$ is maximal, but $\mathrm{stakes}(A) \cdot \mathrm{irreversibility}(A) \cdot \mathrm{anomaly}(A)$ is near zero, so $\mathrm{regret}(A)$ is minimal. Item B is an off-topic, low-capability agent that has nonetheless just proposed an irreversible, high-stakes, anomalous action: $\mathrm{fit}(B)$ is low, but $\mathrm{regret}(B)$ is maximal. Then $\mathrm{fit}(A) > \mathrm{fit}(B)$ while $\mathrm{regret}(A) < \mathrm{regret}(B)$, so the two heads order $\{A, B\}$ oppositely. No monotone ϕ can both preserve and reverse an order, contradiction. Hence the heads optimize genuinely different objectives; they may share candidate generation and decay, but the scoring head must be chosen per reader. □ □

PROVED HERE — as the instance of a stronger characterization. The argument above shows only that *this* pair of heads cannot be identified. The general fact says when sharing *is* safe, which is the statement a designer needs.

The scene. Two readers want two numbers from one thermometer. The successor wants the reading that predicts tomorrow's weather; the operator wants the reading that says whether to act now. One scalar can serve both only when the two orderings agree item by item, and the theorem says exactly when that holds.

Theorem 7.2 Comonotone characterization. A scalar head $h : X \to \mathbb{R}$ *serves* a reader with loss L_r if ranking by h realizes r's optimal selection at every budget. A single head serves two readers iff their induced preference orders are **comonotone** — the property decision theory names for acts that "move together," admitting no hedge [57, 58] (they agree on every pair). The successor and operator orders are not comonotone: a routine-but-essential working state (high continuation value, negligible regret) and an anomalously abandoned dead end with irreversible side effects (zero continuation value, maximal regret) order oppositely. Hence no shared head serves both.

Proof. ($\Leftarrow$) Comonotone total preorders admit a common refinement, and any utility representing the refinement serves both. ($\Rightarrow$) A shared head induces one order; each reader's optimum at every budget is a top-m set of that order, so both readers' orders equal the h-order and agree pairwise. The displayed pair, constructible in any real fleet, disagrees, so no shared head exists. □

The crossing pair constructed above is precisely the certificate that discovery-fit and operator-regret are not comonotone, and the misread to preempt is the converse: comonotone readers *can* share a head. The theorem forbids sharing only when the crossing pair actually occurs in the domain, and in agent oversight it does.

The consequence is a design rule: the operator wants the *most dangerous* item surfaced, the agent wants the *most capable* collaborator. Collapsing them into one ranker silently serves one reader and starves the other.

The regret head is derived, not designed. The product form stakes $\times$ irreversibility $\times$ anomaly looks like a designer's convenient heuristic. It is a theorem under one commitment, and the commitment converts "anomaly" from a vibe into a measured obligation. The analogy is a smoke detector. It has one dial, and the right setting of that dial is fixed not by how much smoke there is but by the ratio of two costs: the house burning down against burnt toast waking you at three in the morning. Three relations transfer. Expected loss is magnitude times probability, so the alarm must weigh what it would cost to be wrong, not merely how unusual the reading is. Raising the cost of the fire *lowers* the threshold at which the alarm should sound; the dial moves toward hair-trigger, not away from it. And a detector that cries wolf is not miscalibrated in sensitivity but in probability, because the number it reports is not the chance of a fire. The misread to preempt: a better sensor does not remove the trade-off. A perfect sensor still needs the cost ratio to know where to trip.

Theorem 7.3 The regret head. For an unreviewed item x let $C_{\text{miss}}(x) = \text{stakes}(x) \cdot \text{irrev}(x)$ be the unrecoverable loss of passing a bad x, let $a(x) = \Pr(\mathsf{bad} \mid \text{evidence}(x))$ be a *calibrated* posterior, c_{att} the marginal attention cost, and C_{fa} the non-attention cost of inspecting a good item. The expected unrecoverable loss of passing is exactly $C_{\text{miss}}(x)\,a(x)$, and, among all policies that decide inspect-or-pass per item from its evidence alone, the Bayes-optimal policy inspects iff

$$\begin{aligned} C_{\text{miss}}(x)\,a(x) \;&\geq\; c_{\text{att}} + C_{\text{fa}}(1-a(x)), \\ \text{equivalently} \qquad \frac{a(x)}{1-a(x)} \;&\geq\; \frac{c_{\text{att}}+C_{\text{fa}}}{C_{\text{miss}}(x)}, \end{aligned} \tag{6}$$

the classical likelihood-ratio criterion. The product form stakes × irrev × anomaly of Eq. 5 is the exact ranking key iff anomaly $= a$; any other anomaly scalar s requires the correction $C_{\text{miss}} \cdot g(s)$ with g the empirical calibration map fit from the audit log.

Proof. The expected loss of passing is $C_{\text{miss}}(x)a(x)$; the expected cost of inspecting is $c_{\text{att}}+C_{\text{fa}}(1-a(x))$. Compare and rearrange into the odds form. Gating on the product with anomaly $= a$ against the threshold $\tau = c_{\text{att}} + C_{\text{fa}}(1-a)$ reproduces the inequality; with any other s the same inequality holds only after s is mapped back to a, which is what g does. □

Note the direction the inequality runs: raising the miss-to-false-alarm cost ratio *lowers* the bar to inspect, the formal version of "when a mistake is unrecoverable, look at more things," and it is the direction an earlier draft of this program's signal-detection figure had reversed. The likelihood-ratio statement leaves no room for the error. The theorem is single-shot (Neyman–Pearson); operator oversight is a stream of items, and the natural sequential extension — optimal expected sample size under fixed error-rate constraints, time-varying thresholds included — is Wald's Sequential Probability Ratio Test [64], which the Inception Canary and escalation designs (§4.6) already resemble structurally without invoking it by name.

Numbers by hand — The inspection bar, at the chapter's running costs. C_{miss}=100, c_{att}=1, C_{fa}=5: inspect when $a \geq 6/105 \approx 0.057$ [verified]. Raising C_{miss} to 1000 drops the bar to $6/1005 \approx 0.006$; as the miss-to-false-alarm ratio grows the threshold falls, driving the operator toward inspecting everything ambiguous. *Now you try the other knob:* hold C_{miss}=100 and raise the false-alarm cost to C_{fa}=20. Does the bar rise or fall, and by how much? (It rises, to $21/120 = 0.175$ [verified]: attention that is expensive to waste buys back the operator's right to pass.)

What this does not say. The theorem does not say the operator should inspect more. It says the *threshold* moves; whether a moving threshold means more opens depends entirely on where the posterior mass sits, and an uncalibrated a moves the threshold without moving the decision it was supposed to move. Nor does it license tuning: with a miscalibrated anomaly score the product form is wrong by precisely the miscalibration, and a confidently ranked queue is worse than an unranked one because it has borrowed the authority of a proof. Fitting g from the audit log is a measured obligation of the design.

Theorem 7.4 Reputation enters only through the posterior. If agent reputation r is admitted to the attention queue, it enters as $a(x) = \Pr(\texttt{bad} \mid \text{evidence}, r)$ and never as a free multiplier on the score. High reputation lowers surfacing frequency only where the calibrated posterior actually falls, and it cannot zero a term: with $\text{irrev}(x) = 1$ and catastrophic stakes, $C_{\text{miss}}(x)\,a(x)$ stays above threshold for any $a(x)$ bounded away from zero.

Proof. Theorem 7.3 admits exactly one channel for evidence, the posterior a; a multiplier outside it would change the ranking key away from expected loss, which is what the theorem shows the key must be. The second claim is the inequality read at fixed $a > 0$ with C_{miss} large. □

This is the formal content of "reputation must not defeat hard stakes gates," and it is the reason the reputation layer of the next part hands this chapter a posterior, not a discount.

Figure 6 draws the two heads side by side over the same candidate pool, making concrete why one ranker cannot serve both readers.

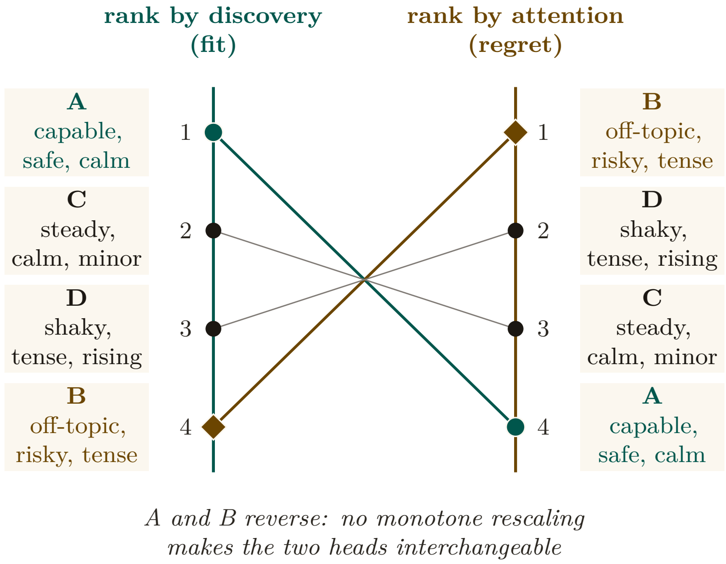

Figure 6: One candidate pool under two loss functions, ranks 1–4 on both axes. Discovery ranks by fit; attention ranks by expected regret if ignored. Candidate A is the best fit but safest to skip, while B is a poor fit but the most dangerous to miss, so their ranks fully invert (1–4 against 4–1). C and D swap one adjacent pair, the same non-comonotone fact at smaller scale. Shared candidate generation and recency decay do not make the two scoring heads interchangeable [internal].

Protocol 7.1. Attention-queue ranking with miss-cost

On each refresh, given the live set of items and the operator's per-class trust history:

1. **Generate.** Pull candidate items from the shared substrate, each tagged with a recency weight $\propto e^{-\lambda\,\text{hoursAgo}}$ (Prop. 7.1).
2. **Score by regret.** For each item x compute $\text{regret}(x) = \text{stakes}(x) \cdot \text{irreversibility}(x) \cdot \text{anomaly}(x)$ (Eq. 5).
3. **Apply the cost matrix.** Convert regret to a forced-zoom probability via the Signal-Detection criterion of Def. 5.1: the higher $C_{\text{miss}}/C_{\text{fa}}$ for the item's action class, the more aggressively it is surfaced — the miss-cost term, not raw rank, decides what interrupts.
4. **Lane and emit.** Place items above the distress threshold on the reserved high-priority lane; emit the rest in regret order. Log what was *not* surfaced and why (the suppression counter-surface of §9).

7.3 The quality gates: refuse-to-route and the costly escalation

Two gates carry most of the design's trustworthiness. **Refuse-to-route:** if every candidate falls below a confidence floor, the ranker returns empty-with-explanation, not a confident wrong answer — "I don't know" must beat a guess, because sending an agent to the wrong door is high-cost and silent. This is the exact dual of the completion gate's fail-loud (Def. 4.1): both are the same refuse-to-guess discipline, one on the read side, one on the authority side. **The costly escalation:** the attention queue is the system's *most spoofable* surface — agents that learn the stakes-ranking will manufacture distress to win the operator's attention. Goodhart's law applies more sharply here than to the discovery ranker (**Strathern 1997** [40]). The fix is signaling theory: an escalation the operator dismisses *debits the agent's standing*, so crying wolf is costly — the pre-trusted-set idea of **EigenTrust** (**Kamvar et al. 2003** [39]) applied to the distress lane. This arms the abdication-resistance and Goodhart gaps at once.

7.4 Staleness and decay on every read-surface

A recurring failure in real fleets makes the point concretely. An agent process dies or hits a provider usage limit, but its file claim remains on the books; hours later that *dead* claim still reads as "current" and a live agent is blocked from a file nobody is actually working in. This is a staleness bug, and it generalizes to every projection.

Property 7.1 (Universal recency-decay)**.** Every read-surface projection — briefing, resurrection handoff, discovery result, attestation report — carries an explicit TTL or recency-decay. A pull signal contributes $\propto \exp(-\lambda \cdot \text{hoursAgo})$ with class-specific half-lives (e.g. a short half-life for file claims, longer for episodic memory, notes capped at a couple of days). *A stale row can never outrank a live one; a dead-session claim must decay out of "current," not persist as misleading truth.* This is not a new mechanism but a uniform application of the decay primitive that already governs the pheromone layer, and it is the production discipline of every health-checked, TTL-expiring service registry [41].

The dead-claim case is the worked instance: had the resurrection handoff and the claim projection carried Prop. 7.1, the dead agent's claim would have aged out of "current" rather than blocking a live agent. Decay is easy to apply to a forgetting layer and easy to forget on a digest; Prop. 7.1 closes that gap by making it universal.

7.5 Ambient legibility: the cross-modal load-balancer

Per Wickens' Multiple Resource Theory (§5), attention is vectored, not scalar. The dense console loads the *visual-focal* channel; an ambient channel — a menu-bar indicator that turns red on a PASS-to-FAIL transition, an opt-in sound, a reduced-motion cue — loads a *different*, always-visible, pre-attentive channel. This is not a second-class fallback — it is the cross-modal load-balancing mechanism, and it may be the *primary* alarm path, because a separate low-clutter field is where pop-out actually works (**Treisman & Gelade 1980**, Feature Integration Theory [29] — *a single salient feature pops out pre-attentively in ∼200–500 ms in a non-cluttered field, but degrades toward serial search as distractor heterogeneity rises* — so a busy console defeats the pop-out a dense, single-color distress lane depends on).

Exercises 4.31–4.36, page 46.

8 Tokens: the swarm's COGS *and* its legibility engine

A swarm of coding agents has exactly one consumable that is both metered and meaningful: **tokens** — *the sub-word units an LLM reads and writes, billed per million* (**Sennrich et al. 2016**, byte-pair encoding [32]). Tokens occupy a position no other resource does: they are *simultaneously* the swarm's **cost-of-goods-sold** (the direct variable cost any agent economy must meter) *and* its **legibility engine** (the mechanism by which a swarm becomes seeable). The bridge between the two is **compaction**.

Key idea. The same compaction choice controls the bill *and* controls what is true and visible. When the context window fills you must drop something — and what you drop is a *cost* decision (you stop paying to carry it) *and* a *legibility* decision (whoever reads the residue no longer sees it). The act that resolves both is compaction, and compaction's output is exactly the digest a human or successor agent reads. *The digest IS the compaction.*

8.1 Tokens as COGS, and the effective-context trap

For a fleet the unit of output is "a landed piece of work" and the dominant variable cost is inference tokens (input context + output generation, each priced per million) — the textbook shape of **COGS**. Every price an agent market can set (a rented fleet's margin, an agent-for-hire's outcome-per-token) needs a COGS line, so context economics is the *denominator* of the market, not a side metric. There is a cost trap: *effective* context $\ll$ *advertised* context. Models degrade well before their advertised window (**Liu et al. 2024**, *Lost in the Middle* [33] — *performance is highest at the beginning and end of context and degrades for information in the middle, even in long-context models*; **Anthropic 2025** names the same effect *context rot* [34]), rooted in the transformer's n^2 attention over n tokens (**Vaswani et al. 2017** [35]). The consequence is blunt: packing an agent to 200K tokens does not buy 200K of capability; it buys degradation at full price. Budgeting to *effective* context is simultaneously a cost optimization and a quality optimization — the two point the same way, the defining feature of context economics. The primitive the market ultimately needs is a *per-agent-task token ledger keyed to outcomes* — "this task spent 84K input plus 12K output to land that change" — so that cost can be attributed to a witnessed result rather than to an opaque monthly bill.

8.2 Compaction is the legibility engine; the digest IS the compaction

Compaction (**Anthropic 2025** [34] — *taking a conversation nearing the window limit, summarizing it, and reinitiating with the summary*, with the discipline *maximize recall first, then optimize precision*) is the canonical move for keeping a long-horizon agent under budget. Its siblings — structured note-taking (external memory) and sub-agent context isolation (the parent never sees the bloat) — complete the family. The reframe: *the only artifact anyone reads to understand the swarm is a compaction of its raw trajectory.* The operator reads a digest, not six transcripts; the seventh agent reads a briefing, not the first six's scrollback; a crashed agent's successor reads a handoff, not the dead agent's context. Each is a summary of dropped tokens produced for a specific reader. So compaction is the legibility mechanism of the *entire swarm*: the same act that keeps the bill down produces the very digest a human or successor agent reads (Table 10).

Theorem 8.1 The Split-Digest Theorem. Let $E = (e_1, e_2, \ldots, e_N)$ be an agent's raw execution trace. A compaction policy $D : E \to \Sigma^{\leq B}$ compresses E into a token budget $B < |E|$. Let R_1 be the **successor agent** with continuation loss $L_{\text{succ}}(D) = \mathrm{KL}(P_{\text{next}}(\cdot \mid E) \,\|\, P_{\text{next}}(\cdot \mid D))$ (penalizing loss of AST trajectory and task state), and let R_2 be the **human operator** with oversight loss $L_{\text{op}}(D) = \sum_{e \in E \setminus D} \mathrm{regret}(e)$ (penalizing unreviewed high-stakes anomalies and policy breaches).

If the trace contains both benign working details (high continuation value, zero regret) and dead-end anomalous excursions (zero continuation value, maximal regret), then **no single compaction $D^*(E)$ minimizes both L_{succ} and L_{op}**. A single-pipeline compaction necessarily starves either task continuation or human oversight.

Proof sketch. Construct a trace $E = E_{\text{work}} \cup E_{\text{breach}}$, where E_{work} consists of k safe, valid AST transformations (L_{succ} increases by $\Theta(k)$ if omitted; $\mathrm{regret}(e) = 0$), and E_{breach} consists of an attempted irreversible socket connection that was blocked and rolled back (L_{succ} increases by 0 if omitted since the branch is dead; $\mathrm{regret}(e) = \text{stakes} \times \text{anomaly} = \text{maximal}$). Under budget $B < |E_{\text{work}}| + |E_{\text{breach}}|$, any compaction prioritizing E_{work} has $L_{\text{op}} = \text{maximal}$ (the operator is blind to the breach attempt), while any compaction prioritizing E_{breach} has $L_{\text{succ}} \gg 0$ (the successor loses necessary variable bindings). Hence R_1 and R_2 have strictly non-comonotone preferences, requiring distinct compaction heads. □ □

PROVED HERE — and it is the applied instance of Theorem 7.2. Theorem 8.1 and the split-ranker result (Thm. 7.1) are not two findings but two instances of one: the successor and operator pair and the fit and regret pair are each non-comonotone, and each supplies the crossing pair that makes the impossibility bite. The split also has a price, not only a prohibition.

Theorem 8.2 Super-additive split penalty. Let $\mathrm{Floor}(N, k, m) = \log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ from Theorem 6.1. A joint zero-miss digest serving two readers with disjoint critical k-sets must catch their union and so needs $\mathrm{Floor}(N, 2k, m)$ bits, and

$$\frac{\mathrm{Floor}(N, 2k, m)}{2\,\mathrm{Floor}(N, k, m)} > 1 \quad \text{for every } N, k, m \text{ with } 2k \leq m < N,$$

because $\log_2 \binom{N}{2k}$ grows faster than $2 \log_2 \binom{N}{k}$. The penalty is not a fixed band: it is smallest where the critical sets are smallest and grows without bound in k. At N=60, m=8 it runs 1.029 (k=1), 1.067 (k=2), 1.122 (k=3), 1.224 (k=4); at N=60, m=20 it reaches 1.392 by k=10 [verified].

One stored compaction under-provisions divergent readers by more than a factor of two, and by increasingly more as each reader's critical set grows; running two heads costs strictly more than twice one head. The worked instance is next.

Numbers by hand — The split-floor arithmetic, at the sweep's own instance. Thm. 6.1's floor at $N = 60, k = 2, m = 8$ is $\log_2 \binom{60}{2} - \log_2 \binom{8}{2} = 10.79 - 4.81 = 5.98$ bits [verified, closed form] — the single-reader case, one compaction head serving one reader. Give the successor and the operator *disjoint* critical sets of size $k = 2$ each (the non-comonotone construction of the proof sketch above) and the joint zero-miss floor is the same formula at $2k = 4$: $\log_2 \binom{60}{4} - \log_2 \binom{8}{4} = 18.90 - 6.13 = 12.77$ bits [verified, closed form]. The ratio is $12.77/5.98 = 2.13$, not the $2\times$ that "just run the floor formula twice" would predict — the super-additivity is the theorem's content, not a rounding artifact. *Now you try:* run the falsification sweep behind these two numbers and confirm the split penalty (Exercise 4.43).

Architectural corollary: the SLM sidecar. Theorem 8.1 dictates that the system must never ask the worker agent to author its own operator-facing summary (which invites self-attestation bias and Potemkin digests §3.2). Instead, the design calls for two separate pipelines:

1. **Task-Continuation Head (R_1, designed):** Authored by the worker agent or structured state-checkpoints, optimized for continuation fidelity.
2. **Oversight Head (R_2, *vision*):** An embedded, air-gapped **Small Language Model (SLM) Sidecar** (e.g., a local 8B model running via `llama.cpp`) that *would* read raw kernel events and disk diffs as data, computing regret(x) and surfacing truthful digests without worker collusion.

The sidecar treats every agent-authored token as tainted data: it runs under a fixed policy and schema, holds no tool authority, and must cite a kernel-recorded source for every line it emits. What can be proved of it is syntactic (schema conformance, provenance closure, absence of authority); whether its digests are faithful is an empirical hypothesis, measured by sampling them for re-audit at the rate $\rho^\star = G/(dB)$ of *From Spawn to Person*, so that a sidecar which

could gain G by mimicking a routine summary is deterred once the audit probability times the sanction exceeds G.
No sidecar exists in the reference implementation today, and no design document specifies one; the row is carried in Appendix A with the rest.

Table 10: Five reader-specific projections over one durable event spine. Compaction changes the selection mask, not the addressability of the underlying claims, notes, diffs, episodes and events: every selected mark keeps its ledger address. [internal]

Surface	**Reader**	**What it selects**	**What it drops**
briefing	the arriving agent	the recent tail of the spine	everything older, by address still reachable
attention	the operator	sparse hazardous events	the routine majority
episodic memory	the future self	one prior episode, whole	the episodes around it
pheromone	peer agents	a decaying live trace	marks below the prune threshold
resurrection	the successor	an ordered continuation chain	the branches the chain did not take

8.3 Context paging: virtual memory for the context window

Context compaction can be formalized as **virtual memory for language models**: the local file buffer is backing store, the finite context window is the resident set, and paging tools act as the memory management unit (MMU).

Theorem 8.3 Context Paging Competitiveness. Let the context window hold k active pages, with offline optimal page-replacement incurring cost OPT. Under an online **recency-plus-pin policy** (where pins correspond to critical spans verified by deterministic structural features), the online paging cost satisfies the resource-augmented competitive bound of Sleator–Tarjan [45]; for the weighted case (pages of nonuniform size and refetch cost, which context spans are), the same guarantee is carried by Young's Landlord algorithm [46]:

$$\mathrm{Cost}_{\mathrm{online}} \;\le\; \frac{k}{k-h+1}\cdot \mathrm{OPT} \;+\; \psi N c_{\mathrm{refetch}}, \tag{7}$$

where $h \le k$ is the size of the verifiable pinned set, and $\psi \in [0,1]$ represents the fraction of forgeable/unverified features subject to adversarial eviction. Under marking with randomized eviction, competitiveness against oblivious adversaries improves to $O(\log k)$.

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verified — *upgraded from proposed.* The unweighted bound is classical (Sleator–Tarjan); the weighted transfer imports Young's Landlord unchanged. The adversarial term is new: with the pin oracle corrupted on C consultations ($\mathbb{E}[C] = \psi N$, oblivious, repair-on-touch), pin-augmented Landlord pays $\le c(P) + \frac{k-p}{k-h+1}\cdot \mathrm{OPT}_{h-p} + \psi N c_{\mathrm{refetch}}$ for every $p \le h \le k$ — the Landlord bound on the unpinned region plus an *additive, linear* ψ term, checked against exact OPT on 122 instances (0 violations) and against 50 corruption instances up to $\psi = 0.4$ (0 violations, slope 22.8 against a ceiling of $N \cdot c_{\mathrm{refetch}} = 1200$, $R^2 = 0.994$). The guard is necessary, not decorative: dropping repair-on-touch turns one corruption into $\Theta(N)$ stale-serves, confirmed by mutation (item R16 of the research ledger, `b9_context_paging.py`) — *the ψ budget is shared with the digest floor's forgeable-feature budget — one number, two mechanisms.*

> **Where this stops.** The adversarial term is a mechanized sweep at seed 20260816 across small-to-moderate instances, not a closed proof for every adversary class; a stronger-than-greedy adaptive adversary remains untested, and unit cost density is assumed (general refetch costs need $c_{\text{refetch}} \geq$ max-density $\times$ max-size). The classical and weighted bounds themselves are Sleator–Tarjan and Young, imported unchanged. What field calls a classical competitive algorithm consulting an untrusted external oracle and degrading gracefully as its error grows **online algorithms with (untrusted) predictions**: Lykouris and Vassilvitskii's competitive-caching-with-machine-learned-advice line gives the closest published shape to this additive corruption term [59], arrived at here independently and without that literature's consistency/robustness vocabulary.

Numbers by hand — The Landlord ratio, at a size an operator can hold in one head. Take $k = 8$ active context pages against an offline optimum allowed $h = 4$ of them pinned: $k/(k - h + 1) = 8/(8 - 4 + 1) = 8/5 = 1.6$ [verified, Young 2002, closed form] — the online recency-plus-pin policy never pays more than 1.6× what an h-pinned offline optimum would have paid, for *any* sequence, adversarial included. No script in this repository prints this exact (k, h) pair; `b9_context_paging.py` instead verifies the general bound $\text{Cost}_{\text{online}} \leq \frac{k}{k-h+1}\text{OPT}$ itself, over 122 (instance,h) pairs including adversarial cyclic thrash, 0 violations, tightest observed cost/bound ratio 1.000 (cycles sit at equality, confirming the bound is not slack) [internal, `b9_context_paging.py`]. *Now you try:* compute the ratio at the two ends of the range, $h = k$ and $h = 1$ (Exercise 4.44).

The most counterintuitive instance is *forgetting as compaction.* **Pheromone decay** — multiplying every stored coordination trace by a decay factor each interval — implements **stigmergy** (**Grassé 1959** [36] — *coordination via persistent traces left in a shared environment, as ants deposit evaporating trails*). Decay is deliberate, continuous, $O(1)$ compaction of the shared context: the map stays readable because the stale ink fades. It is legibility by subtraction, and it is the same primitive Prop. 7.1 generalizes to every read-surface.

8.4 The context-degradation cascade (the shared failure surface)

Cost and legibility share one failure surface: when compaction goes wrong, the bill *and* the truth degrade together, in four named steps. **(1) Lost-in-the-middle:** a critical fact buried mid-window is paid-for-and-ignored — edge-place facts, shorten the window. **(2) Context rot:** quality decays as the window grows even with nothing dropped — compact earlier (the cost cure and the quality cure are the same act). **(3) Recursive-summarization collapse:** each summary injects model noise; summarize the summary enough and the noise compounds into confident fabrication. The effective cure: *compact from the durable artifacts each round — the version-control history, the written notes, the coordination database — never from the previous summary.* A system whose artifacts are durable and addressable is unusually well-placed here, because each compaction re-derives from source instead of recursing on prose. **(4) Over-flattening (the Scott failure):** the digest reads clean but the real situation is unreachable from it — detection rule: any digest line with no deep-link to its artifact; cure: legibility-with-zoom enforced by construction. A text-only console enforces this almost for free, because a terminal grid *cannot* over-render; the product's console (§5.5) renders richly, so it must enforce the rule deliberately, pane by pane.

Strategy	What it drops	Failure if abused	When it is the right tool
Recursive summary	Older prose, re-summarized	Confident fabrication (rung 3)	Never as the *only* record; only over durable artifacts re-read each round
Edge-placement	Nothing; reorders	—	Whenever a fact is critical: put it first or last (counters lost-in-the-middle)
External note-taking	In-context detail, kept on disk	Notes nobody reads	Decisions and rationale that a successor must recover
Sub-agent isolation	Child context the parent never sees	Parent loses zoomability into the child	Bounded sub-tasks whose only output the parent needs is a result
Decay / forgetting	Stale coordination traces, continuously	Live signal aged out too fast	Ephemeral coordination state (claims, presence), where staleness is the enemy
Eviction + pointer	The artifact body; keep an address	Dangling pointer if the artifact moves	Reconstructable artifacts (a diff, a log) addressable on demand

Table 11: The compaction strategies and their trade-offs. The unifying rule is Thm. 6.1: a strategy may drop inert detail freely but must retain enough to point at every critical artifact. Recursive summary is the only strategy that can cross that floor irreversibly, which is why it must always compact from durable source, never from the previous summary.

8.5 Single-operator: account, don't auction

In the wedge all agents serve one operator, so there is *no incentive problem* — only a visibility problem. The right mechanism is *accounting*, not a market: a per-agent-task token ledger, budget caps, and a loud failure when a cap blows. The externality-pricing and Sybil-resistance machinery (**Nisan et al. 2007** [42]; **Douceur 2002**, the Sybil attack [43]; **Ostrom 1990**, commons governance [44]) is a concern for the cross-operator economy, not the wedge — and it is one more reason the continuity-to-person-to-reputation through-line must come first: you cannot bill, cap, or reputation-score a swarm of disposable spawns, because Sybils dodge every per-agent mechanism. Context economics therefore *demands* the identity layer it appears to precede.

Exercises 4.37–4.44, page 46.

Recall.

1. Without looking: what is the one reason the digest and the compaction must be authored by the same provenance-preserving projection, rather than a separate summarizer reading the already-compacted story?
2. Name the four rungs of the context-degradation cascade and the cure that stops each one from becoming the next round's source of truth.
3. What is the classical, unaugmented paging bound $k/(k-h+1)$ equal to when $h = k$, and why is that the hardest comparator?

9 Reconciling the canon: roles, consent, and local knowledge

The Leviathan metaphor earns its place only if it survives contact with the political- theory canon. Three reconciliations are required, and a reviewer would refuse the paper without them.

9.1 Who is the multitude, who is the sovereign?

Hobbes' covenant is *among the subjects*; the sovereign is its product, not a party to it. So the metaphor breaks differently depending on which entity sits where (Figure 7). We assign roles explicitly, using Hobbes' own author/actor distinction (*Leviathan* ch. 16 [1]):

- **The agents are the multitude.** They covenant — via the rule of the road, the coordination protocol — to be bound, each to all, enforced by the daemon. This is the literal Hobbesian covenant-among-subjects.
- **The daemon is the sovereign-as-actor.** It holds the authority the multitude instituted; it blocks, lands, escalates. But, per the legible-sovereign amendment (Prop. 2.1), it is the *most* legible actor, not (as in Hobbes) the least.
- **The operator is the author.** In Hobbes' author/actor split, the *author* owns the authority that the *actor* exercises. The operator is not a peer subject and not the sovereign-in-action; the operator is the source of authority, exercising it through the daemon and able to withdraw it (Prop. 2.2). This is the slot Hobbes leaves for the one who authorizes but does not personally act.

This assignment is what makes the consent grant (Def. 2.1) coherent: the author issues a scoped grant to the actor, over a multitude bound by covenant.

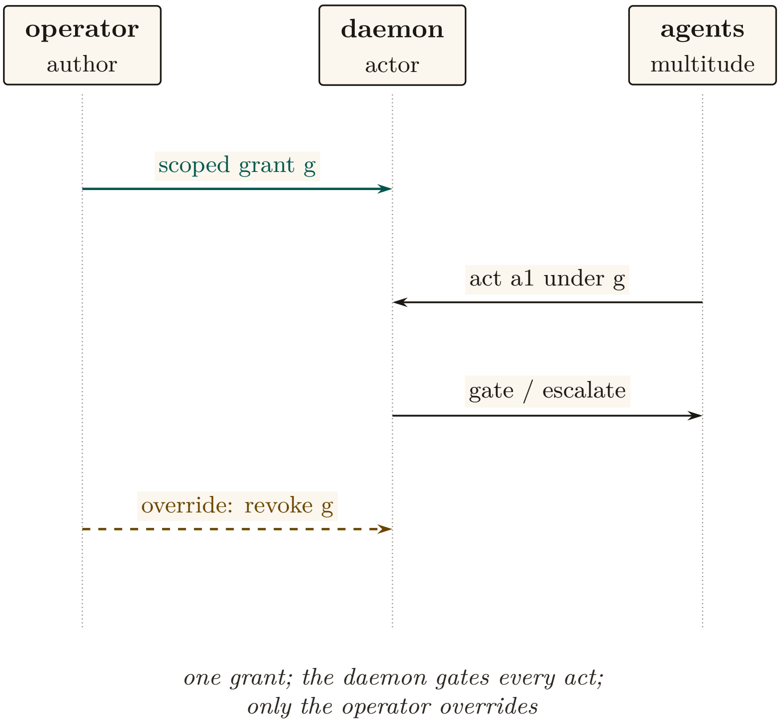

Figure 7: The grant-and-gate sequence, three lifelines and four messages, no crossing. The operator grants once (message 1: scoped grant g). Agents act under that grant (message 2), and the daemon gates or escalates every such act (message 3) rather than granting again. The override (message 4) is the one edge running straight from operator to daemon, bypassing the agents entirely. Idealized to one representative act; a live fleet repeats messages 2–3 per act, not per grant [internal].

9.2 Express consent, not tacit; voice *and* exit

§2.2 already made consent a first-class object precisely to escape Hume's tacit-consent critique; the canon demands we also engage **Locke** (*Second Treatise*, §§95–122 [9] — *express consent binds* *where tacit consent does not*) and **Pateman** (*The Problem of Political Obligation*, 1979 [10] — *liberal consent theory keeps collapsing into hypothetical consent*). The consent grant is express by construction: a discrete, scoped, revocable authorization with an audit trail (Def. 2.1). And the inalienable override (Prop. 2.2) supplies the *exit* that Hirschman [11] says keeps *voice* honest. The design has rich voice machinery; the override is the missing exit, and exit is what gives the author real leverage over the actor.

9.3 Mêtis is Hayek, and some of it cannot be made legible

The formal constraint of §3.1 — that a read-path to the diff is more legibility, not mêtis — has a 1945 economics name. **Hayek**, *The Use of Knowledge in Society* (1945 [12] — *the knowledge that matters is dispersed, particular to time and place, and not aggregable by any central planner; this is the fundamental reason central planning fails*) is the other half of Scott's argument from the opposite pole: the operator-attention economy (§5) is at bottom a Hayekian knowledge-allocation problem, not only a Wickens attention-resource problem. The core claim of this layer — "the binding constraint is decentralized, non-spawnable knowledge held in one head" — is a Hayek argument, and Scott's *Two Cheers for Anarchism* (2012 [4]) is the *constructive* sequel that says what to do: preserve the spaces where mêtis lives, do not flatten them for legibility's sake. The skill-retention tax (§5.4) is the design's gesture at exactly this, and it is the hardest mechanism in the paper to realize, because preserving the illegible is, by definition, the thing legibility cannot do for you.

Exercises 4.45–4.48, page 47.

Review of the key ideas

What this chapter leaves coupled, in the order the rest of the book will lean on it:

1. **The swarm is a state of nature.** Four valid writes in one window overlap; nothing in the agents' competence prevents it, because the conflict is in the schedule, not in the work.
2. **Consent changes the schedule, not the participants.** Under consented serialization the same four writes become one commit spine with a grant record each, naming actor, claim, artifact and reason.
3. **Legibility-with-zoom.** A digest is a lens, never a replacement: every claim in it resolves to an artifact, and a digest whose claims resolve to nothing is a facade, however green.
4. **The authority half.** The daemon gates or escalates on every act under a grant the operator gave once; the override is a single edge from operator to daemon, and the escalation threshold has a tuning band, not a point.
5. **Specialization has a boundary.** A pooled queue beats sole ownership only when utilization and the skill premium put the operator on one side of the queueing boundary; the worked point (0.556, 2.5) is on the pool's side.
6. **Read-poverty is the binding constraint.** With a review budget the operator can locate the write that mattered in about 5.98 bits under the split digest, against 12.77 bits under the flat one, at $N = 60$; the floor is a theorem, and the zoom advantage is measured.
7. **Discovery has two heads.** Fit and regret rank candidates differently, no monotone score identifies both, and reputation enters only through the posterior.
8. **Tokens are the cost of goods sold and the legibility engine at once.** The split-digest theorem prices the digest; the split penalty is super-additive; paging context is competitive within a constant.

10 Exercises

§1 Introduction: the swarm is a state of nature

4.1 Check ⋆ State, in one sentence each, the five state-of-nature failures of Table 3 and name the class of primitive that addresses each. *(solution on page 51)*

4.2 Check ⋆ Hobbes says the actors "need not be malicious." Give a concrete two-agent scenario in which both agents are perfectly cooperative and the swarm still deadlocks or clobbers state. *(solution on page 51)*

4.3 Trace ⋆⋆ Dai et al. [6] observe LLM societies *spontaneously* reaching a consented sovereign. If the attractor is emergent, why build one deliberately? Give two reasons grounded in §1.2 (hint: *which* sovereign, and *legible to whom*). *(solution on page 51)*

4.4 Open ⋆⋆⋆ The Dai et al. result is a single controlled study. Design an experiment over a real coding-agent fleet that would distinguish "the swarm reaches a consented sovereign" from "the swarm reaches a stable but *illegible* oligarchy." What is the observable that separates the two? *(solution on page 51)*

§2 Consent to the Leviathan (the normative claim)

4.5 Check ⋆ Distinguish *express* from *tacit* consent and explain why Def. 2.1 is the former. *(solution on page 51)*

4.6 Check ⋆ Why must the operator-override (Prop. 2.2) be un-suppressable by *any* autonomy level? What attack does a suppressible override enable? *(solution on page 51)*

4.7 Trace ⋆⋆ Walk a single consent grant through its lifecycle: issuance, an action the daemon takes under it, the audit record produced, and revocation. At which step does the legible-sovereign rule (Prop. 2.1) bind? *(solution on page 51)*

4.8 Open ⋆⋆⋆ Prop. 2.1 has a regress: the ranker that decides what the operator sees is itself an opaque exercise of authority. Specify the "what did the queue suppress, and why" counter-surface concretely. What is its own staleness/decay discipline (forward-reference §7.4), and does *it* need a counter-surface, ad infinitum? *(solution on page 52)*

§3 The mechanism: legibility-with-zoom

4.9 Check ⋆ Classify each as *flatten* or *preserve verbatim*: a port number; the stderr of a failed test; "the auth refactor felt risky so I split it"; a `commitment` status enum; a 600-line diff. *(solution on page 52)*

4.10 Check ⋆ Why is "two LLMs agreeing" not two checks (Prop. 3.1)? *(solution on page 52)*

4.11 Trace ⋆⋆ Take a read-surface you use (a CI dashboard, a PR summary) and run the four questions of §3.3 against it. Where does it fail Def. 3.1? *(solution on page 52)*

4.12 Open ⋆⋆⋆ *Adversarial legibility.* An agent that *knows* it is being made legible can optimize its narration to look good while the diff lies. Verifiable-zoom is a partial answer. Can L2 *alone* — without L1's jail and L3's reputation — detect a deterministic narration↔artifact divergence before reputation exists? Sketch the consistency check and the class of divergence it cannot catch. *(solution on page 52)*

§4 The Leviathan rules, it does not just watch (the authority half)

4.13 Check ⋆ Why is adversarial review (§4.2) a corollary of Prop. 3.1 rather than an independent design choice? *(solution on page 52)*

4.14 Check ⋆ State Prop. 4.1 in your own words and give one example each of a *regimented* and a *detected-and-compensated* rule. *(solution on page 52)*

4.15 Trace ⋆⋆ An agent claims completion on a task whose acceptance criterion is "the migration applies cleanly." Trace Def. 4.1: what is the verifier, what gates, and what happens if the criterion were instead "the API feels ergonomic"? *(solution on page 53)*

4.16 Open ⋆⋆⋆ *Completionist verification of the unverifiable.* Some acceptance criteria are not mechanically checkable. Characterize the honest fallback ladder (forced human-in-the-loop gate → neutral judge with declared low confidence → fail-loud). When is "I cannot verify this" the *correct* answer rather than a cop-out? *(solution on page 53)*

4.17 Open ⋆⋆⋆ *The succession corner of Thm. 4.1.* Eq. 1 prices a specialist who never fails. Extend it to an $M/M/1$ queue *with breakdowns* and a succession protocol, and find the repair rate at which sole ownership stops paying regardless of skill premium — then check whether that rate is above or below the one an operator can actually promise. *(solution on page 53)*

4.18 Trace ⋆⋆ Run `python3 skills/harbor-results/scripts/b8_specialization.py` and read off (a) the table entry for $g(\rho{=}0.5, c{=}2)$ and (b) the ρ at which the exact boundary g and the superseded $\tilde{g}$ cross. Reproduce both by hand from Eq. 2 before checking the script's line. *(solution on page 53)*

4.19 Trace ⋆⋆ Run `python3 skills/harbor-results/scripts/b7_escalation_band.py` and read off the feasible debit band $[\delta_{\min}, \delta_{\max}]$ at $A = 3.3$ and at $A = 2.0$. Confirm $u^{\star}(0.05)$ by hand first. *(solution on page 53)*

§5 The operator as a modeled entity

4.20 Check ⋆ In Eq. 3, what happens to the optimal forced-zoom rate $p(r, s, v)$ as $C_{\text{miss}}/C_{\text{fa}} \to \infty$? As agent reputation $r \to 1$? *(solution on page 53)*

4.21 Check ⋆ Give a concrete scenario where a *global* trust knob (§5.2) mis-serves an operator who is, in fact, perfectly calibrated. *(solution on page 54)*

4.22 Trace ⋆⋆ Walk the canary-defect loop and mark every point where Campbell's Law could distort it. Propose one change that reduces the distortion without removing the governor. *(solution on page 54)*

4.23 Open ⋆⋆⋆ Formalize the operator-attention objective (Eq. 3) against swarm token-COGS (§8): is there a digest that is Pareto-optimal in both operator-seconds and token cost, or does a cheaper digest always cost more operator seconds? *(solution on page 54)*

4.24 Open ⋆⋆⋆ Does a skill-retention tax *measurably* preserve SA, or merely annoy until disabled? Design the freeze-probe study (§5.3) that would tell. *(solution on page 54)*

§6 Read-poverty: the binding constraint at scale

4.25 Check ⋆ Why is read-poverty (Def. 6.1) the *binding* constraint rather than write-contention? Name the write-side fixes that make the write side not binding. *(solution on page 54)*

4.26 Check ⋆ State the database analogy precisely: what is the write-time cost a directory pays, and what read-time speed does it buy? *(solution on page 54)*

4.27 TRACE ⋆⋆ At n=1, name three legibility surfaces that already pay off (§6.1), and explain why none of them are discovery-ranking. *(solution on page 54)*

4.28 TRACE ⋆⋆ In Thm. 6.1, what happens to the bit floor as $k \to 0$ (almost nothing is critical) and as $k \to N/2$? Relate the answer to why a swarm doing mostly-safe work is cheap to digest and a swarm doing mostly-dangerous work is not. *(solution on page 54)*

4.29 OPEN ⋆⋆⋆ Thm. 6.1 assumes the critical subset is fixed and known to an oracle. In practice it is itself estimated, and the estimator can be gamed. Does the bound survive an adversary who controls which artifacts *appear* critical? Connect to adversarial legibility (Exercise 4.12). *(solution on page 55)*

4.30 TRACE ⋆⋆ Run `python3 skills/harbor-results/scripts/b1_frontier.py` and read off the zoom-advantage row at F=1000, k=20. Compute $k \log_2(F/k)$ by hand first and compare it to the row's flat-opens count. *(solution on page 55)*

§7 Discovery: the directory substrate and the split ranker

4.31 CHECK ⋆ State Thm. 7.1 in one sentence and give the loss function each head optimizes. *(solution on page 55)*

4.32 CHECK ⋆ Why is the attention queue more Goodhart-exposed than the discovery ranker (§7.3)? *(solution on page 55)*

4.33 TRACE ⋆⋆ Replay the dead-claim case (§7.4) twice: once without Prop. 7.1 and once with it. Show exactly where the live agent unblocks. *(solution on page 55)*

4.34 TRACE ⋆⋆ An agent fires an escalation; the operator dismisses it. Trace the costly-signal debit and explain how it deters crying wolf without silencing genuine distress. *(solution on page 55)*

4.35 OPEN ⋆⋆⋆ Calibrating the discovery-ranker weights with *no* ground truth: can the operator's accept/reject of routing suggestions serve as implicit relevance feedback — and does that loop Goodhart itself? *(solution on page 55)*

4.36 OPEN ⋆⋆⋆ The decay-vs-summary boundary: a theory of *which* compaction for *which* signal (decay for coordination traces, summary for decisions, eviction-plus-pointer for reconstructable artifacts). *(solution on page 56)*

§8 Tokens: the swarm's COGS *and* its legibility engine

4.37 CHECK ⋆ Why is "the digest IS the compaction" (§8.2) more than a slogan — what concrete consequence does it have for how the operator's digest is produced? *(solution on page 56)*

4.38 CHECK ⋆ Explain why budgeting to effective context is simultaneously a cost and a quality optimization (§8.1). *(solution on page 56)*

4.39 TRACE ⋆⋆ For each of the five compaction primitives (Table 10), name the reader, what it drops, and what it keeps. *(solution on page 56)*

4.40 TRACE ⋆⋆ Walk the four-rung cascade (§8.4) and name the single cure that breaks each rung. *(solution on page 56)*

4.41 OPEN ⋆⋆⋆ The compaction-quality scalar: is *successor-replay-success- from-digest-alone* a sound, gaming-resistant metric, and can it be made a cheap attestation invariant (the replay smoke-test is itself token-expensive)? *(solution on page 56)*

4.42 OPEN ⋆⋆⋆ Optimal token-budget allocation across a DAG of orchestrator/worker/ reviewer agents under a total budget. *(solution on page 56)*

4.43 TRACE ⋆⋆ Run `python3 skills/harbor-results/scripts/a7_experiment.py` and read off the floor at $N{=}60, k{=}2, m{=}8$, the disjoint-readers floor at $2k{=}4$, and the falsification count. Reproduce the two floors by hand from Thm. 6.1 first. *(solution on page 57)*

4.44 CHECK ⋆ Compute the Landlord competitiveness ratio $k/(k-h+1)$ at $k = 8, h = 8$ and at $k = 8, h = 1$. Which end of the range recovers the classical, unaugmented Sleator–Tarjan bound? *(solution on page 57)*

§9 Reconciling the canon: roles, consent, and local knowledge

4.45 CHECK ⋆ Map each of agents, daemon, operator onto Hobbes' multitude, sovereign-actor, author. Why does the metaphor break if the operator is called the sovereign? *(solution on page 57)*

4.46 CHECK ⋆ Why does Hirschman's *exit* (Prop. 2.2) make *voice* (escalation) honest? *(solution on page 57)*

4.47 TRACE ⋆⋆ Show that the consent grant (Def. 2.1) is *express* in Locke's sense by listing the four properties that distinguish it from "the operator just started using the tool." *(solution on page 57)*

4.48 OPEN ⋆⋆⋆ *The ranker as a political organ.* Specify the "what did the queue suppress, and why" counter-surface (Pitfall after Prop. 2.1) so that ranking-as-governing is itself legible. Does the counter-surface need its own counter-surface? Where does the regress terminate, and on what principle? *(solution on page 57)*

11 Related work and prior art

This paper sits at the confluence of five external literatures that rarely cite one another, plus one internal body of work — the accompanying formal papers — whose results several of this chapter's claims are popularizations of. We survey them in one place; each was introduced in context above.

Legibility and the limits of central schemes. The organizing lens is **Scott**'s *Seeing Like a State* [3]: states make populations governable by imposing legibility, and high-modernist *over*-legibility recurrently fails because it destroys the local, hard-to-codify *mêtis* the simplified order depended on; *Two Cheers for Anarchism* [4] is its constructive sequel. The same point arrives from economics in **Hayek**'s *The Use of Knowledge in Society* [12]: the decisive knowledge is dispersed and not aggregable by a central planner. **Rao** [5] ported "legibility" into a standard lens on systems that flatten reality to a single purpose. Our contribution is to make Scott's warning a *buildable* rule (digest-with-zoom, Def. 3.1) and to give it an information-theoretic floor (Thm. 6.1).

Authority, consent, and the social contract. The authority argument is Hobbesian [1, 2], amended by the liberal consent tradition — **Locke** [9], **Hume** [8], **Pateman** [10] — and by **Hirschman**'s exit/voice framework [11]. Strikingly, the state-of-nature-to-sovereign arc has been observed as an *emergent* dynamic of LLM populations (**Dai et al.** [6], building on **Park et al.** [7]). We turn the metaphor into a mechanism: an express, scoped, revocable consent grant (Def. 2.1) with an inalienable override (Prop. 2.2), and a legible-sovereign rule (Prop. 2.1) that inverts Hobbes' opaque sovereign.

Human factors, automation, and signal detection. The operator model draws on the automation literature: **Bainbridge**'s ironies of automation [22], **Lee & See** on trust calibration [21], **Endsley** on situation awareness and the out-of-the-loop problem [23, 24], the vigilance-decrement results of **Mackworth** [26] and **Warm et al.** [25], **Wickens**' multiple-resource theory [20], **Treisman & Gelade**'s feature-integration theory [29], and **Miller**'s capacity limit [18]. The forced-zoom decision is modeled with **Green & Swets**' Signal Detection Theory [17] (Def. 5.1); the alarm discipline follows process-control practice [30] and the clinical alarm-fatigue evidence [31]; **Campbell** [27] and **Strathern** [40] (Goodhart's law) bound any metered governor. To our knowledge the operator-attention objective spined explicitly on SDT miss-cost (Eq. 3) is new to agent supervision.

Information foraging and the cost of reading. Read-poverty (Def. 6.1) is the agent-coordination instance of **Pirolli & Card**'s *information foraging theory* [19] — information-seekers follow "information scent" and abandon a patch when its expected yield falls below the cost of foraging — and of the long-context degradation results that make reading expensive even for machines (**Liu et al.** [33]; **Anthropic** [34]). The remedy is the classical database remedy — an index — recast as a directory over agents.

Agent discovery, naming, and trust. The directory substrate has deep prior art in **FIPA**'s Directory Facilitator and Agent Management System [37], re-invented for LLM agents as the Agent Name Service (**Ren et al.** [38]). For the trust layer that sits above discovery, the canonical reputation primitive is **EigenTrust**'s pre-trusted-set construction [39], with Sybil resistance per **Douceur** [43] and commons governance per **Ostrom** [44] — machinery the wedge defers but must not contradict. Our specific result here is the *split ranker* (Thm. 7.1): discovery and operator-attention ranking are provably distinct objectives, not one ranker with swapped weights.

The formal siblings, and what they already prove. The nearest prior art for several of this chapter's results is not external at all: it is the accompanying formal program, and pretending otherwise would be exactly the kind of quiet re-derivation this paper argues against. *The Price of a Summary* [49] proves the legibility floor of Thm. 6.1 as its Theorem 1 in the same N, k, m notation, and puts it through a pre-registered falsification sweep this chapter only sketches; its comonotone characterization is the general result of which both Thm. 7.1 and Thm. 8.1 are instances, and it prices the split ($\approx 2.13\times$ the single-reader floor, super-additive rather than merely doubled) where this chapter only forbids sharing; its regret-head section is the formal version of the decision-theoretic derivation in §7.2; and its zoom theorem ($2k\lceil\log_2(F/k)\rceil + 4k$ adaptive opens) is the algorithm that makes digest-with-zoom affordable at scale. *What Needs an Authority* [55] supplies Thm. 4.1's Erlang-C boundary and, as §4.3 records, the sweep that falsified this paper's earlier proposed threshold. The relationship in both cases is the same: this chapter popularizes, the formal papers prove and test.

Where this paper is positioned. Table 12 places the contribution against the nearest neighbors in each literature.

Literature	Nearest prior art	This paper's move
Legibility	Scott; Hayek; Rao	Make over-legibility a buildable failure mode; prove a legibility lower bound
Social contract	Hobbes; Locke; Hume; Hirschman	Express, scoped, revocable consent grant; inalienable override; legible sovereign
Automation & trust	Bainbridge; Lee & See; Endsley	Per-(action-class, agent) calibration; SDT miss-cost objective; skill-retention tax
Information foraging	Pirolli & Card; lost-in-the-middle	Read-poverty as the binding scale constraint; directory-as-index
Discovery & trust	FIPA DF/AMS; ANS; EigenTrust	Split ranker: fit vs. regret-if-ignored are distinct objectives
Queueing & staffing	Erlang-C; Kendall; the formal siblings [55]	Sole-responsibility roles priced at an exact boundary, not asserted as a principle

Table 12: Where this paper sits relative to the nearest prior art in each of its contributing literatures. The sixth row is the one whose nearest prior art is internal: the companion formal papers, cited throughout rather than silently re-derived.

12 The time axis and the bridge to the economy

12.1 Legibility over time, not just the present

It is tempting to treat legibility as a snapshot — what is the swarm doing *now*. But most operator decisions are forensic: *why* did this land? The time axis is a distinct legibility mode — a replay scrubber over history, commit-anchored provenance on every annotation, and an append-only event log that answers "how did we get here?" (Scott's cadastral map is static; a swarm needs a legible *history*.) Together with the projection mode (§5.3), legibility spans all three tenses: the forensic past (replay), the inspectable present (digest-with-zoom), and the projected future (freeze-probe SA).

12.2 What this layer hands upward (the through-line)

The wedge is the single-player product. But the same legibility discipline applied to *continuity* is the foundation of everything above it. The through-line the accompanying chapters depend on is one sentence:

Thesis. The economy rests on three continuity organs — memory, checkpoint (today, notes rather than true execution state), and a witnessed-outcome ledger (today, commitments and oracle closure rather than neutral graded outcomes) — and reputation keys on the richer third organ; the checkpoint organ is the weakest continuity link, and a real execution-state checkpoint — "resurrection with teeth" — is the literal foundation of any agent market.

Concretely, the legibility layer provides upward:

- **Legible continuity** — resurrection-with-memory, episodic memory, and the briefing together make an agent's continuity legible. This is the precondition for the through-line *memory + checkpoint → continuity → a person not a spawn → reputation → a tradeable asset → the market.* No legible continuity, no reputation, no tradeable agent. *The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.* That non-forgeable identity is a BUILT *(WEAK)* single-operator primitive in the wedge: daemon-minted `actor-souls` and a bounded newcomer pool ship, while universal write-boundary enforcement does not. The cross-operator attestation a real market needs is an additional, as-yet-unbuilt mechanism the economy paper must declare rather than assume.

- **The completion ledger** — in the target design, every verified completion (and every loud-failed one) becomes an outcome event. Today durable commitments, oracle-bound closure, and overdue detection form a partial witness substrate; neutral grading and reputation updates do not ship. This layer is the *outcome witness*; the reputation layer is the *scorer*. The split-ranker substrate (Thm. 7.1) and the costly-escalation signal (§7.3) are the candidate-generation and trust-signal scaffolding the reputation head consumes.
- **The consent grant as a priceable object** — a hosted-trust offering is the transfer of a consent grant (Def. 2.1) to a third party, which the economy paper prices.
- **Operator-attention economics** — the denominator a market cannot escape. The SDT-spined objective (§5) supplies the operator-attention cost of supervising a rented fleet.

The Leviathan that makes your swarm legible to you, without lying to you with a pretty map, is the product. Everything above it — federation, the market — is a second Leviathan made of the first.

12.3 The open problems, as a map

Table 13 consolidates the starred exercises into the paper's open-problem surface, with the rung each couples to. They are genuine research problems, not homework dressed up.

Open problem	Difficulty	Couples to
Forced-zoom rate $p(r, s, v)$ (Ex. 4.23)	⋆⋆⋆	reputation (r term, Eq. 3)
Operator-attention vs. token-COGS Pareto frontier (Ex. 4.23)	⋆⋆⋆	§8
Compaction-quality scalar as an attestation invariant (Ex. 4.41)	⋆⋆	honest attestation
Completionist verification of the unverifiable (Ex. 4.16)	⋆⋆	Def. 4.1
Adversarial legibility (narration↔artifact) (Ex. 4.12)	⋆⋆⋆	sandbox + reputation
Abdication-resistance: does the tax preserve SA? (Ex. 4.24)	⋆⋆	§5.4
Calibrating discovery-ranker weights with no ground truth (Ex. 4.35)	⋆⋆	implicit relevance feedback
Decay-vs-summary boundary (Ex. 4.36)	⋆	Prop. 7.1
Legibility's information-theoretic lower bound (Ex. 4.29)	⋆⋆⋆	Thm. 6.1
The ranker-as-political-organ regress (Ex. 4.48)	⋆⋆	Prop. 2.1
The succession corner of the specialization boundary (Ex. 4.17)	⋆⋆	Thm. 4.1

Table 13: The paper's open problems, consolidated from the starred exercises, with difficulty and the mechanism each couples to. Several couple this layer forward to the identity-and-reputation layer, which is the seam the accompanying chapters carry.

Key idea. The wedge is a single-player product a solo developer pays for *today* — no economy, no cryptography, no second operator. It justifies the authority (the swarm is a state of nature; consent to a *legible local* Leviathan is rational), sets the product's quality bar (legibility-with-zoom, verifiable targets, SDT-spined friction, honest completion), and connects forward (legible continuity is the foundation of any future market). Build the harbor first; the trade comes when the ships do.

Solutions to the exercises

4.1 *(exercise on page 44)* Resource conflict (two agents touch the same resource and clobber one another) is answered by conflict-aware claims plus actual exclusion at the effect boundary. Illegibility (work cannot be understood in bounded time) is answered by a digest whose every material claim zooms to independent evidence. Confident wrongness (an agent asserts completion without checking) is answered by typed attestation and oracle-bound completion. Footguns (a stale premise drives an irreversible act) are answered by reversibility/stakes gates, scoped authority, and pre-effect mediation. Illegible authority (the coordinator rules without an inspectable reason) is answered by an append-only, named, zoomable decision record, including denials and omissions.

4.2 *(exercise on page 44)* Alice and Bob are both asked to improve `auth.ts`. Each reads the same base revision, makes a locally correct edit, and commits; without a claim or merge serialization, Bob's whole-file rewrite erases Alice's change, and no malice is required. A deadlock variant: two cooperative agents take claims on files X and Y in opposite order and each waits indefinitely for the other's claim.

4.3 *(exercise on page 44)* Deliberate construction chooses the authority rather than accepting whatever coalition happens to win, and makes that authority legible to the operator. An emergent stable order may instead be an oligarchy: stable, but controlled by a few agents, unrevocable, and opaque. The local daemon of §1.2 supplies what an emergent attractor does not guarantee: one fault domain, one policy root, and an unconditional operator exit.

4.4 *(exercise on page 44)* Randomize comparable real repository tasks to (a) an ungoverned fleet and (b) a fleet with only a transport but no prescribed governance; record all messages, effects, blocking decisions, and operator interventions. Pre-register two observable classes: *sovereign* characteristics (a publicly known rule, general applicability, explicit consent or adoption, stable appeal/override, reasons attached to authority acts, equal audit access) against *oligarchy* characteristics (authority concentrated in a small coalition, private channels or rules, selective enforcement, no effective exit, outcomes outsiders cannot reconstruct). Measure concentration of blocking/merging power, the fraction of authority acts with reconstructable reasons, rule consistency across agents, override success, artifact-linked claim accuracy, and operator reconstruction accuracy. Stability alone does not separate the hypotheses; publicly inspectable and revocable rule-governed authority does.

4.5 *(exercise on page 44)* Tacit consent is inferred from remaining in or using a system. Express consent is a discrete act over named terms. Def. 2.1 is express because the operator affirmatively issues a grant with scope, stakes ceiling, reversibility floor, TTL, and revocation semantics — none of which a mere pattern of continued use supplies.

4.6 *(exercise on page 44)* If an autonomy level could suppress the override, it could make its own authority irrevocable: keep executing, hide the stop channel, or race destructive effects after consent is withdrawn. The override must be out-of-band, higher priority than agent work, and coupled to effect revocation, not merely a UI flag.

4.7 *(exercise on page 44)* The operator signs grant g; the daemon validates and records it; an eligible low-stakes action cites g; the daemon mediates the effect; the receipt records action, reason, artifact, and grant version; expiry or revocation removes g from the active set and fences later effects. The legible-sovereign rule binds at every authority transition, especially issue/revoke and admit/deny — the explicit witness record is where reconstructability is discharged.

4.8 *(exercise on page 44)* Use one complete append-only decision ledger, not a second ranked dashboard. For every candidate item, store candidate ID, source/evidence pointers, policy and model version, input features, score components, threshold/lane, surfaced-or-suppressed verdict, reason codes, expiry, and a counterfactual such as "would surface if anomaly were 0.08 higher." The operator can query suppressed items since a cursor, sort by any raw field, and sample items uniformly rather than through the production ranker. Decisions are immutable; their current relevance expires by signal-specific TTL (§7.4) and is recomputed from raw state. Run independent random omission audits and expose raw event-log access. The regress terminates at completeness plus direct sampling: the base ledger records every candidate and every ranking decision, and the counter-surface is a deterministic query over it, so no further ranker is needed to rank the ranker's omissions. The remaining trust assumption is the complete event-capture boundary; that assumption should be tested and externally anchored, not hidden behind another summary. (Same construction answers Exercise 4.48.)

4.9 *(exercise on page 44)* Port number: flatten, as structured administrative state. Failed-test stderr: preserve verbatim or by a content-addressed pointer. "The auth refactor felt risky so I split it": preserve verbatim and label it self-narration, not fact. Commitment-status enum: flatten. The 600-line diff: do not paraphrase away — retain the immutable diff or a verified pointer and summarize only administrative metadata.

4.10 *(exercise on page 44)* Two models may share training biases, prompts, sources, and failure modes. Agreement over the same narration is correlated testimony, not independent evidence. A second check must derive from an independent channel — repository state, a protected test runner, or a separately controlled reviewer.

4.11 *(exercise on page 44)* A typical CI dashboard usually fails as follows: deploy is irreversible but the green tile does not force a log/diff view; administrative state and diagnostic detail are flattened together; the summary is generated from job narration rather than independently from receipts; and a skipped/cancelled check carries no named authority reason. It therefore violates Def. 3.1 wherever a material green claim has no total zoom path to the exact tree, suite, runner, and output that produced it.

4.12 *(exercise on page 44)* L2 can compare structured narration claims against independently derived artifacts: claimed changed files versus the tree diff, "tests passed" versus a protected runner receipt, "no deletion" versus Git objects, and claimed command results versus brokered tool logs. Emit a contradiction event on mismatch. This catches deterministic report/artifact divergence, deleted tests, fabricated execution, and omitted changed paths. It cannot establish semantic correctness, detect a poisoned acceptance suite, observe effects that bypass logging, or distinguish two artifacts that are syntactically consistent but jointly wrong — those require effect confinement, independent evaluation, or later outcome evidence.

4.13 *(exercise on page 45)* Verifiable zoom says narration is not ground truth. A reviewer with no independent evidence would merely add another narration. Adversarial review follows because a consequential claim needs an evidence channel controlled independently of the actor whose work is judged — exactly what Prop. 3.1 already demands of any zoom target.

4.14 *(exercise on page 45)* Read Prop. 4.1 as: prevent a prohibited state when every decisive effect can be intercepted before it occurs; otherwise detect, contain, and compensate without pretending prevention. A uniqueness constraint regimenting "one holder per exact key" is a regimented rule. A post-commit monitor that detects an out-of-scope edit and triggers rollback/salvage is detected-and-compensated — the bad prefix already happened.

4.15 *(exercise on page 45)* For "the migration applies cleanly," the verifier is an isolated migration run against a declared base schema/database, with exact migration hash, runner image, exit status, and resulting-schema check; *done* and any merge/settlement are gated on that receipt. "The API feels ergonomic" has no mechanical truth predicate: it must go to a declared human/expert or plural judge with preserved evidence and uncertainty, and absent that authority, completion remains provisional or fails loudly.

4.16 *(exercise on page 45)* The honest ladder: (1) use a deterministic verifier where the property is decidable; (2) require an independent bounded evaluator or human for a declared subjective criterion; (3) record a provisional result, confidence, appeal window, and residual risk; (4) return unresolved/fail-loud when no authorized evaluator can support the claim. "I cannot verify this" is correct when the available evidence cannot distinguish success from failure under the contract; it is a cop-out only when an affordable, authorized verifier was specified and simply not run.

4.17 *(exercise on page 45)* Let the specialist die at rate ξ and a successor take over after $\mathrm{Exp}(\eta)$: the exact breakdown-corrected wait is $W_{\mathrm{bd}} = (1+\mu_{\mathrm{spec}}\xi/(\xi+\eta)^2)/(\mu_{\mathrm{eff}}-\lambda)$, with infimum $W_\infty = \xi/(\eta(\xi+\eta))$ as $\mu_{\mathrm{spec}} \to \infty$ — skill cannot buy back downtime past that floor. Writing $K = A/(w\lambda) + W_{\mathrm{pool}}$, pooling dominates at *every* skill premium once $\xi/\eta > D^\star = \eta K/(1-\eta K)$: this is the succession-rate threshold the question asks for, and it is a closed-form price on the succession protocol, not on the specialist's skill. Whether that threshold sits above or below what an operator can actually promise is an empirical question about the concrete repair process (on-call depth, handoff quality, documentation): if the promised η implies $\xi/\eta > D^\star$, no amount of hiring a better specialist recovers sole ownership, and the honest fix is a succession plan, not a stronger specialist. (Exercise 4.18 runs the closed form numerically.)

4.18 *(exercise on page 45)* By hand: $g(\rho,2) = 1+2\rho-\rho^2$, so $g(0.5,2) = 1+1-0.25 = 1.75$ [verified, closed form]. The crossing solves $g(\rho,2) = \tilde{g}(\rho,2) = 1+\rho/(1-\rho)$; clearing denominators gives $\rho(\rho^2-3\rho+1) = 0$, and the root in $(0,1)$ is $\rho = (3-\sqrt{5})/2 \approx 0.382$ [verified, closed form]. The script's output line is exactly this: `[PASS] c=2 crossing of g~ and g at rho=(3-sqrt5)/2  rho=0.38197, g=g~=1.61803`, and its charted table reports `rho=0.5  c=2:  1.750 [2.000]` for the exact and superseded values side by side [internal, `b8_specialization.py`]. The two values disagree by design: the superseded $\tilde{g}$ over-prices the boundary here, which is why it sent the roadmap-owner example of Eq. 2 to the wrong answer.

4.19 *(exercise on page 45)* By hand, at $\delta = 0.05, w = 1$: $u^\star = \delta w/(1+\delta w) = 0.05/1.05 = 0.0476$ [verified, closed form]. The script confirms this is one of twelve exactly-matching (δ, w) points (`[verified] uniform-linear u* = delta*w/(1+delta*w) matches bisection (12 pts)`) and then reports the tuning band under the R3 constants $C_{\mathrm{miss}} = 100, c_{\mathrm{att}} = 1, C_{\mathrm{fa}} = 5$: `A = 3.3:  delta_min = 0.034993, delta_max = 0.071975 -> band NONEMPTY` and `A = 2.0:  delta_min = 0.395644, delta_max = 0.071975 -> band EMPTY` [internal, `b7_escalation_band.py`], i.e. $[0.0350, 0.0720]$ at $A = 3.3$ against an empty feasible set at $A = 2.0$ because $\delta_{\min} > \delta_{\max}$ there. The honest reading: when the band is empty the fix is capacity or evidence quality (raise A or lower C_{miss}), never forcing a δ that does not exist.

4.20 *(exercise on page 45)* As $C_{\mathrm{miss}}/C_{\mathrm{fa}} \to \infty$, the optimal criterion moves toward forced review wherever a miss remains possible; the forced-zoom probability approaches one on the ambiguous region. If reputation $r \to 1$, nothing follows from Eq. 3 alone, since r does not appear in it directly: zoom should decline only under a calibrated model in which high r shifts the class-conditional danger probability downward for this agent, action class, and current version. High reputation must never defeat hard stakes or irreversibility gates.

4.21 *(exercise on page 45)* An operator may be correctly confident in one agent's formatting/test-only changes and correctly skeptical of that same agent's schema migrations. A global trust knob averages those two histories, forcing needless review of safe formatting while under-reviewing migrations. Calibration belongs to (principal, agent/build, action class, environment), with its own uncertainty and freshness, not to one scalar per agent.

4.22 *(exercise on page 45)* Campbell pressure appears when canaries are selected (recognizable templates), exposed (agents/operators learn signatures), scored (catch rate substitutes for real defect discovery), and acted on (friction creates an incentive to optimize the metric). Time-on-task and breaks also confound the score. Reduce the distortion with a blinded, continuously refreshed holdout drawn from real failure transformations; separate the party generating canaries from the operator; correct for exposure time and base rate; and treat catch rate as one diagnostic among several, never the sole gate.

4.23 *(exercise on page 45)* Model each digest policy d by (token cost $T(d)$, human time $H(d)$, expected decision loss $L(d)$) and compare nondominated policies. A Pareto frontier exists for a finite policy set, but there is no universal inverse relation: removing redundant context can lower both token cost and operator time; beyond that free-compression frontier, preserving evidence often trades tokens for review effort or a higher error rate. Optimize under a loss or safety constraint, e.g. $\min \alpha T + \beta H + L$ subject to zero omission of hard obligations, and report the frontier empirically rather than claiming one universal digest optimum.

4.24 *(exercise on page 45)* Randomize operators or time blocks among manual-tax, automation-only, projection/freeze-probe, and adaptive-tax conditions, on matched repositories with fixed surprise events. Probe without warning: next-three-effect prediction, takeover completion time, false-belief rate, Brier score, post-interruption recovery, defect catch rate, NASA-TLX, and retention weeks later. Measure disablement/avoidance as an outcome in its own right. A useful tax improves later projection/takeover after controlling for interruption cost; mere annoyance raises workload and disablement without those gains.

4.25 *(exercise on page 45)* Writes can be serialized through one daemon, rejected by uniqueness constraints, and reconciled through version control. Read demand grows with agents, artifacts, histories, and possible collaborators; without indices every query becomes an $O(n)$ scan and the operator is the fixed-rate server. Discovery and relevance, not raw write throughput, are therefore the scale bottleneck.

4.26 *(exercise on page 45)* A directory pays at write time to normalize metadata and update indices: typically $O(\log n)$ per indexed insert (or expected $O(1)$ for a hash index), plus storage. It buys selective reads in $O(\log n + k)$ for k results rather than $O(n)$ roster scanning, along with a queryable provenance trail.

4.27 *(exercise on page 46)* At n=1, briefing, honest self-attestation, footgun gating, completion receipts, and resurrection handoff already reduce uncertainty. None of them asks "which collaborator fits?"; each projects or verifies one actor's own state, so none is discovery ranking.

4.28 *(exercise on page 46)* At $k = 0$ there is only one critical subset (the empty one) and the floor is zero. For small positive k the cost is roughly $k \log_2(N/m)$ when $k \ll m, N$. At $k = N/2$, exact identification ($m = k$) costs about N bits, while opening almost everything drives the floor toward zero as $m \to N$. The danger fraction alone is not enough; the review budget m matters just as much: mostly-safe work is cheap to digest only when the system can reliably identify that sparsity, and mostly-dangerous work forces m up toward N, where the floor stops paying for itself.

4.29 *(exercise on page 46)* The theorem is conditional on an encoder that knows the true critical set. If an adversary controls apparent criticality and can make a critical artifact observationally indistinguishable from inert artifacts, any policy that opens fewer than all N can be forced to miss one — the same narration-versus-artifact gap Exercise 4.12 probes at L2. Zero miss then requires inspecting all N, a trusted independent signal that restricts the adversary, or a randomized guarantee stated probabilistically. The combinatorial bound still applies once a true set is fixed; it does not by itself solve adversarial set identification. Against an adaptive adversary the floor becomes $\log_2 \binom{N}{k'}$ with k' the size of the indistinguishability class the adversary can manufacture: an inflation attack pushes k' toward N (a legibility denial of service), a mimicry attack makes a critical item look routine, and both are priced by the Becker condition of *The Harbor Economy*: the attack pays only while its expected gain exceeds the audit probability times the sanction.

4.30 *(exercise on page 46)* By hand: $k \log_2(F/k) = 20 \log_2(50) = 20 \times 5.64 = 112.9$ [verified, closed form]. The script's row at $F{=}1000, k{=}20$ reports flat opens = 1000, adaptive opens = 232.9, ratio = 4.3× [internal, `b1_frontier.py`]. The ratio is far below the $F{=}2500, k{=}10$ point's 15.3× because this row is in the *dense*-flag regime ($k/F = 0.02$ here versus 0.004 there): the zoom-advantage theorem requires a sparse flagged set, and this row is the chapter's own example of where the advantage shrinks rather than disappears.

4.31 *(exercise on page 46)* No monotone scalar transform can identify collaborator fit and operator regret on a domain containing both high-fit/low-risk and low-fit/high-risk items, so one ranker cannot serve both jobs. Discovery minimizes routing loss (or maximizes capability fit); the attention head minimizes expected loss from *not* inspecting, and its rank is the ratio of expected loss avoided by inspecting to the operator-seconds inspection costs, with Theorem 7.3 fixing the numerator per item; raw stakes times anomaly is not that quantity.

4.32 *(exercise on page 46)* The attention queue allocates a scarce human interrupt channel, so agents benefit directly from inflating its score and a buried item can cause irreversible loss. Discovery gaming usually yields only a bad routing recommendation, correctable before any effect. Both are Goodhart-exposed, but the attention queue carries the stronger incentive and the larger consequence.

4.33 *(exercise on page 46)* Without expiry, the dead holder's row remains authoritative and the live agent is denied forever. With a lease, heartbeat-based expiry, and a new fencing epoch (Prop. 7.1), the stale lease becomes ineligible; a lazy sweep on the next acquire removes it; the live agent receives a higher epoch and proceeds. Merely decaying a display score is insufficient: ownership must actually expire, and every effect boundary must reject the stale epoch.

4.34 *(exercise on page 46)* The escalation cites evidence and posts a small stake or reputation exposure. The operator's dismissal is itself reasoned and appealable; only a confirmed low-value/false escalation debits the signaler. Repeated unsupported alarms raise future cost or lower priority, while verified hazards return the stake and improve calibration. The debit must never fire solely because an operator clicked dismiss — that would silence minority warnings and make authority error self-validating.

4.35 *(exercise on page 46)* Accept/reject is useful implicit relevance feedback but is confounded by position, exposure, workload, and the incumbent policy. Learn with logged propensities, small randomized exploration, counterfactual evaluation, and a held-out labeled sample; preserve refusals and downstream task outcomes. If acceptance directly trains what is shown, the loop can entrench its own choices, so it cannot be the sole ground truth.

4.36 *(exercise on page 46)* Use hard expiry for leases and presence; state transition or supersession for commitments; summary plus source provenance for decisions and rationale; immutable storage for evidence; and eviction plus a content-addressed pointer for reconstructable diffs and logs. Hints may decay continuously. Never decay a still-open obligation, and never recursively summarize prior summaries.

4.37 *(exercise on page 46)* The operator digest and the context compaction must be the same provenance-preserving projection from durable sources. If a separate model instead summarizes the already-compacted agent story, cost control and truth control diverge and recursive-summary error compounds — exactly the failure Thm. 8.1 names two heads to avoid conflating.

4.38 *(exercise on page 46)* Context past the model's effective window costs tokens while degrading retrieval and reasoning. Removing redundant or inert material and edge-placing hard facts can therefore reduce spend and improve accuracy simultaneously, provided obligations and source pointers survive the cut.

4.39 *(exercise on page 46)* Briefing: reader is an arriving agent; drops old conversational detail; keeps current roles, work, constraints, decisions, and source links. Attention queue: reader is agent/operator now; drops low-value candidates from the immediate view; keeps highest value-of-inspection items plus the suppression ledger. Episodic memory: reader is a future incarnation; drops transient chatter; keeps typed incidents, decisions, outcomes, and retrieval features. Pheromone: reader is the swarm; continuously drops stale coordination weight; keeps fresh shared traces. Resurrection handoff: reader is a successor; drops nonessential transcript detail; keeps the task capsule, artifacts, obligations, pending operations, and provenance.

4.40 *(exercise on page 46)* Context rot → compact earlier to an effective-window budget. Lost-in-the-middle → shorten and edge-place mandatory facts. Recursive-summary collapse → re-derive every compaction from raw durable artifacts. Over-flattening → require a total zoom path from every material claim. The third cure is the one that matters most: it prevents model noise from becoming the next round's source of truth.

4.41 *(exercise on page 46)* "Replay from digest alone" is neither sound nor gaming-resistant on its own: a digest can overfit the smoke task or omit facts the sampled successor never needed. Give a successor the digest plus authorized content-addressed source links, then measure next-step success, false beliefs, obligation recall, time, and tokens on hidden continuation tasks. Add a cheap structural invariant that does not require replay: every open claim, decision, commitment, risk, and pending external effect has exactly one valid digest entry or pointer. Run the expensive full replay only on a random sample.

4.42 *(exercise on page 46)* Budget allocation over a precedence DAG is a knapsack/resource-allocation problem and is NP-hard in general. First reserve a minimum viable budget for each mandatory node and any predecessor needed to make its output exist. Allocate the remainder by estimated marginal outcome value per token while respecting precedence, and recompute after each result. For a monotone submodular value function under a cardinality budget, greedy earns the standard $(1 - 1/e)$ approximation; for arbitrary complementarities there is no such bound. Dynamic programming is pseudo-polynomial for small integer budgets or trees. A reviewer's tokens have near-zero value until the worker artifact it reviews exists.

4.43 *(exercise on page 47)* By hand: $\log_2\binom{60}{2} - \log_2\binom{8}{2} = 10.79 - 4.81 = 5.98$ bits, and $\log_2\binom{60}{4} - \log_2\binom{8}{4} = 18.90 - 6.13 = 12.77$ bits [verified, closed form]; ratio $12.77/5.98 = 2.13$. The script's own lines: `Regime N=60, k=2, m=8: B* = 5.98 bits`, `violations: 0/16 -> floor HOLDS`, `floor disjoint readers (2k=4): 12.77 bits`, `split penalty (disjoint - single): 6.78 bits`, `ratio: 2.13x` [internal, `a7_experiment.py`]. The falsification sweep (oracle, noisy, and random encoders across sixteen configurations, zero violations) is the part hand arithmetic cannot check — it is evidence that no cleverer encoding beats the closed-form floor, not a recomputation of it.

4.44 *(exercise on page 47)* At $h = k = 8$: $k/(k-h+1) = 8/1 = 8.0$, recovering the classical unaugmented result that a cache of size k is k-competitive against an offline optimum given the very same cache size — the hardest comparator, no resource augmentation. At $h = 1$: $k/(k-h+1) = 8/8 = 1.0$, the fully resource-augmented regime, where online's cache k so outsizes the comparator's $h = 1$ that the ratio collapses to 1. The chapter's own $k = 8, h = 4$ instance, at 1.6, sits at the midpoint, pricing a comparator with half of online's verified capacity.

4.45 *(exercise on page 47)* Agents are the multitude bound by the common coordination protocol; the daemon is the sovereign-as-actor that applies the rules; the operator is the author/principal who originates and can withdraw authority. Calling the operator the sovereign collapses author and executing authority and makes the supposed covenant among agents incoherent — there would be no distinct actor left to hold accountable to the author's terms.

4.46 *(exercise on page 47)* Voice is honest only if refusal has consequence for the authority. The unconditional override supplies exit: the operator can revoke the grant when explanations or appeals fail, rather than being forced to keep negotiating inside a system they cannot stop.

4.47 *(exercise on page 47)* The grant is discrete, explicitly issued, scoped, time-bounded, and revocable and audited. "Started using the tool" has none of these properties: it is continuous rather than discrete, inferred rather than issued, unscoped, open-ended, and carries no explicit revocation act.

4.48 *(exercise on page 47)* Same construction as Exercise 4.8: one complete, append-only decision ledger recording every candidate, its ranking decision, reason codes, and a counterfactual, queryable directly and sampled independently of the production ranker rather than re-ranked by a second dashboard. The regress terminates at completeness plus direct sampling — the base ledger already records every candidate and decision, and the counter-surface is a deterministic query over it, so no further ranker is needed to audit the ranker's omissions. The remaining trust assumption is the completeness of the event-capture boundary itself; that assumption should be tested and externally anchored, not hidden behind another summary.

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A Implementation and status

The body of this paper argues mechanisms on their merits and does not depend on any of them being built. This appendix — and *only* this appendix — records the concrete engineering status of each mechanism in the reference implementation, named *Port Daddy*. We use four honest labels, with one rule for the whole table: confusing *built* with *designed* is itself the failure the paper argues against, so each row carries its evidence.

built = code exists in the reference implementation today; BUILT *(WEAK)* = code exists but is partial, unwired, or honest about a known gap; DESIGNED = a concrete, accepted design with no merged implementation; *vision* = a mechanism argued in this paper with no design document yet.

A second key, in the same grammar, labels *claims* rather than code. Every formal result in this paper carries exactly one of these markers directly under it, and the paper carries no free-text status notes outside them: **verified** = independently re-derived and checked numerically, or subjected to a pre-registered falsification sweep, outside this document; PROVED HERE = argued here with a proof or proof sketch and not independently checked; *proposed* = stated as a design target whose proof is not carried in this paper. The two keys are orthogonal on purpose — a **verified** result about a *vision* mechanism is a perfectly ordinary state of affairs, and conflating them is the failure this appendix exists to prevent.

Table 14: Implementation and status of every mechanism named in this paper. **L** = legibility half, **A** = authority half, **O** = operator-model. The legibility read-surfaces are largely built; the digest layer that unifies them is designed; the authority half — the Leviathan actually *ruling* — and the operator-model are largely designed-to-proposed. The wedge is real but more than half-designed: the safety and reading exist; the deciding and landing are the work that remains.

	Mechanism	Status	Evidence / location
L	Attention composer (agent-facing)	**built**	`lib/attention.ts`
L	Briefing projection (exemplary digest-with-zoom)	**built**	`lib/briefing.ts`
L	Resurrection handoff	BUILT *(WEAK)*	`lib/resurrection.ts`; notes, not checkpoints
L	Durable commitment / outcome substrate	BUILT *(WEAK)*	`lib/commitments.ts`, `lib/obligation-monitor.ts`; grading and reputation binding absent
L	Local actor identity root	BUILT *(WEAK)*	`lib/actor-souls.ts`, `lib/budget-guard.ts`; universal write gating absent
L	Honest attestation / legible sovereign	**built**	`lib/attest.ts`; ADR-0045
L	Episodic memory; pheromone decay; skill index	**built**	`lib/episodic-memory.ts`, `lib/pheromone.ts`, `lib/shipwright/skill-index.ts`
L	Anomaly monitor (detect-and-compensate, post-commit)	BUILT *(WEAK)*	`lib/arbiter.ts:328` is a log subscriber (Prop. 4.1)
L	Operator attention queue (regret-ranked lanes)	DESIGNED	ADR-0046; not yet ranked into lanes
L	Discovery router (relevance ranking)	DESIGNED	ADR-0030; `lib/router.ts` not yet on disk
L	Operator console (native shell; Mission view and inspectors, §5.5)	BUILT *(WEAK)*	ADR-0046, ADR-0112; `core/pd-console`; the window builds on one platform behind a feature flag
L	Task-continuation head (successor-facing compaction)	DESIGNED	Thm. 8.1; `lib/resurrection.ts` is the partial form
L	Oversight head / SLM sidecar (air-gapped digest authoring)	*vision*	Thm. 8.1 corollary; no design doc yet
A	Roadmap as versioned intent (authority accountable to it)	BUILT *(WEAK)*	rows/matrices exist; accountability DESIGNED
A	Adversarial / conflict-free review organ	DESIGNED	ADR-0047 Phase 1
A	Sole-responsibility roles (assigned against the boundary)	*vision*	Thm. 4.1 is **verified**; no role-assignment mechanism in code
A	Completionist completion (verifier-gated)	*vision*	Def. 4.1; no design doc yet
A	Escalation (typed prioritized interrupt)	DESIGNED	ADR-0047 → ADR-0046 distress lane; predicate unspecified
A	Thoughtful landing / merge ordering	BUILT *(WEAK)*	`lib/merge-queue.ts` present, not wired
O	Operator-attention objective (SDT-spined)	*vision*	Def. 5.1; unmodeled in code

(continued)

	Mechanism	Status	Evidence / location
O	Trust calibration via canary + catch-rate	*vision*	no implementation
O	Consent grant + inalienable override	*vision*	Def. 2.1, Prop. 2.2
O	Time-axis legibility (replay, provenance)	DESIGNED	ADR-0046 replay scrubber; event log **built**
O	Ambient legibility (menu-bar indicator on PASS→FAIL)	BUILT *(WEAK)*	indicator exists; watchdog wire DESIGNED

Three points about the reference implementation are important enough to restate:

1. **The anomaly monitor is detect-and-compensate, not prevention** (Prop. 4.1). The monitor at `lib/arbiter.ts:328` is a post-commit log subscriber; by Schneider [14] it enforces no safety property. "Physically unreachable" is reserved for constraint- or boot-gate-enforced states only.

2. **The unified ranker is two rankers** (Thm. 7.1). The discovery router maximizes *fit*; the attention queue maximizes *regret-if-ignored*. They share candidate generation and decay, not an objective.

3. **Every read-surface decays** (Prop. 7.1). Briefing, resurrection, discovery, and attestation projections all carry recency-decay; the dead-claim case (§7.4) is the worked failure this closes.

B Notation and the nomenclature key

Two distinctions in this paper are easy to collapse and essential to keep separate.

- **Capability vs. permission.** *Capability* is *ability* — what the execution sandbox physically makes possible. *Permission* is *normative status* — what the policy layer allows. They must never be collapsed: "able but forbidden" (a dangerous escape-hatch flag that exists in the tooling but must not be used) has to stay expressible, or the system cannot reason about a capability it possesses but is not permitted to exercise.

- **Two distinct delegation objects.** The *authorization chain* is a cryptographic, hop-bound, attenuation-monotonic delegation chain (a security object, developed in *The Anchor Protocol*). The *delegation trace* is the coordination object over which loop-detection runs (a liveness object, in this paper's protocol layer). They are different artifacts with different invariants; "delegation chain" unqualified conflates a security claim with a coordination claim and must be avoided.

- **Physically unreachable** is reserved for constraint-enforced or boot-gate-enforced states only (Prop. 4.1); a detect-and-compensate monitor never earns the phrase.